



Getting Started with FSx for ONTAP

January 2026 | LD01279 Version 3.0.1

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1 Introduction

Existing native AWS file storage services such as Amazon Elastic File Service (Amazon EFS) and FSx for Windows File Server have provided a good starting point for the basic cloud storage needs of a business, focusing on single protocols and specific operating systems or workloads. Because of their simplicity and turn-key operation, managed storage services like these allow companies to be more agile and scalable with their cloud infrastructure.

However, in shifting to managed AWS storage, AWS customers lose the disaster recovery and protocol flexibility provided by NetApp ONTAP storage. Additionally, when making this transition, companies are left without features that improve storage efficiency such as deduplication and data tiering. Some companies need these features but wish to maintain the convenience and cost structure of using a managed storage service.

FSx for ONTAP is a native, fully managed AWS storage service solution for these customers. The service provides the operational benefits of AWS managed storage and includes many of the familiar features found in on-premises ONTAP storage offerings.

The service can be used for network file shares, cloud-based backup and disaster recovery, primary applications, cloud-native applications, and speeding up dev/test cycles. It is also the only fully managed AWS service to offer multi-protocol capabilities for Windows and Linux workloads.

FSx for ONTAP services can be deployed and managed using either the AWS console, or NetApp Workload Factory. NetApp Workload Factory can be accessed from the NetApp Console, which is the NetApp centralized data management platform for both on-premises and cloud storage. NetApp Console brings ease of deployment and management to advanced ONTAP features like SnapMirror, FlexCache, and FlexClone.

From NetApp Console, users can take advantage of NetApp Workload Factory if they are also a hybrid cloud customer using AWS and FSxN for NetApp ONTAP.

- Some of the advantages of NetApp Workload Factory with NetApp Console includes:
 - It's free to add with no additional costs. Just create an account and add your AWS credentials to get started.
 - Offers data protection visibility across workloads.
 - NetApp Console specific benefits include the following:
 - Manage ONTAP in any cloud from one console experience while adding management features not available from the AWS Console alone.
 - System Manager-like UI
 - Data tiering management
 - Replication
 - NetApp Workload Factory specific benefits:
 - Helps you find opportunities to manage growing costs better and implement to best practices.
 - The Codebox feature lets you accelerate infrastructure-as-code for multiple formats: API requests, CloudFormation, and Terraform.
 - More Snapshot policy control than what's available in AWS Console.
 - Multi-protocol file sharing access.
 - Simplified Block storage management.
 - Access to FlexClone technology.
 - Automatic Capacity Management.
- Access to try and subscribe for Data Services:
 - NetApp Backup and Recovery
 - NetApp Ransomware Resilience
 - NetApp Data Classification
 - NetApp Disaster Recovery (Great for VMware Cloud on AWS customers with FSx for NetApp ONTAP and VMware running ONTAP anywhere).

In this lab, you use NetApp Console to deploy and manage an FSx for ONTAP environment. After deploying the service you will configure some of the supported protocols and provision a new volume. Then you will explore some of the management features available to you, all within NetApp Console.

As you go through this lab, keep in mind that because FSx for ONTAP is powered by ONTAP, you are able to take advantage of the complete portfolio of NetApp products and services (such as Cloud Sync, Cloud Data Sense, and Global File Cache), and manage all of these integrations in one place with NetApp Console.

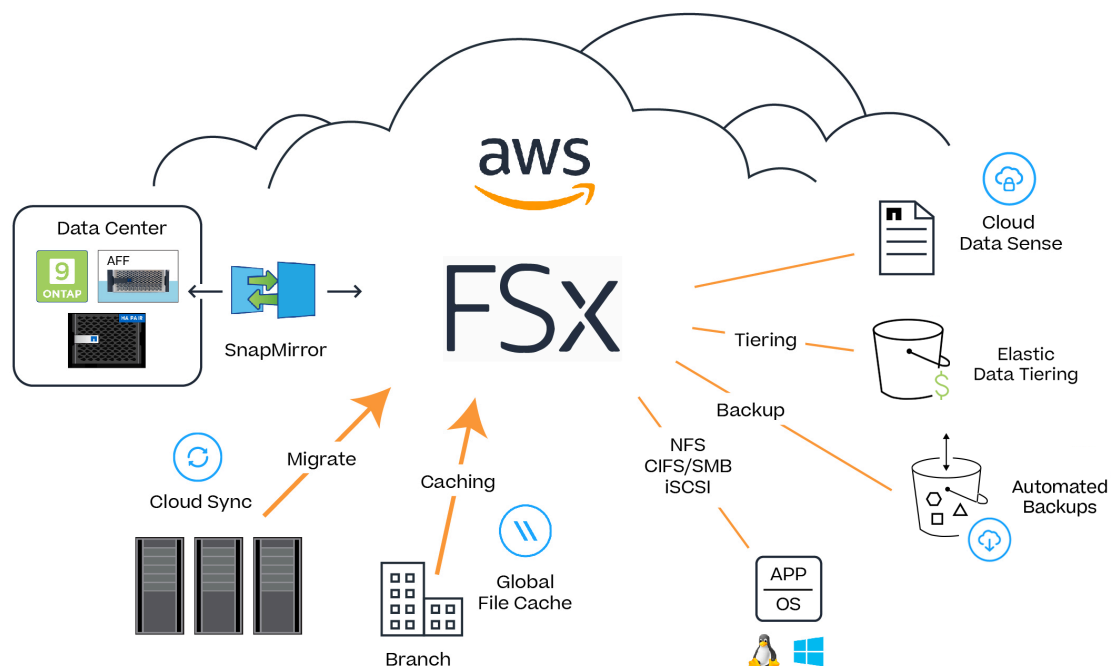


Figure 1-1:

Note: This lab uses a simulated environment to demonstrate some of the capabilities of the NetApp Console. Some portions of this demonstration use pre-populated sample data while others directly reflect the tasks you perform through the course of the lab. Feel free to explore this simulator beyond the scope of this guide; you will not break anything.

2 Lab Activities

In this lab you will explore the following topics:

- [Prepare Your Lab Environment](#)
 - [NetApp Console Standard Mode Access](#)
- [Setting Up FSx for ONTAP](#)
- [Provision and Manage Amazon FSx for NetApp ONTAP Volumes](#)
 - [Provision a Volume with Storage Efficiency and Tiering](#)
 - [Explore Data Management Features](#)
- [Data Replication with Amazon FSx for NetApp ONTAP](#)

2.1 Prepare Your Lab Environment

In this section, you log into the lab environment and ensure that the automation is complete, and then obtain your credentials and log into a NetApp Console Organization that has pre-created for this experience.

 **Important:** The steps performed in this section are required to prepare the simulated demo environment used in this lab, but the user interface you interact with is the real hosted service platform.

This section includes the following activities:

Activities
NetApp Console Standard Mode Access

2.1.1 NetApp Console Standard Mode Access

In this activity, you navigate to the in-lab account information where you can then use those supplied credentials to log into the NetApp Console. This lab has a dedicated NetApp Console Organization, Project, and Agent that were created before you connect to the lab. To facilitate quicker access to the demonstration, the account(s) for use with NetApp Console IAM have been pre-created.

1. Click the automation status link on the desktop (most times it will be a green check, but if action is required it may be a yellow warning, and instructions are provided after clicking).
2. You are on <http://ae.demo.netapp.com/status>. Click **Accounts** which takes you to <http://ae.demo.netapp.com/accountsiam>.
3. Look for the user labeled “Main user.” If other users are present, you may use those in later sections of this lab guide. Since only one user can be logged into the NetApp Console at a time, you may need to either log out of the Main user browser session, or if you want to be logged in with two different credentials you can use the browser “private” or “incognito” browser tab feature.
4. Click the link to the right of that NetApp Console section (the box with arrow pointing up) to go to the console.

- Copy, then paste both the Username and Password from the Main user on the Accounts page to your lab session's browser to log into the pre-created Organization in the NetApp Console.

NetApp Lab on Demand | Lab Status | Accounts **2**

Accounts

rt11324NTB ()

LOD Cloud Demo environments

Accounts

NetApp Console Simulator

Environment Information	Organization Name	Project Name
	LOD_rt11324NTB	rt11324NTB 3
Main user Role: Organization admin	Username	Password
	rt11324NTB@netapp-lab.com	 5

4

Figure 2-1:

2.2 Setting Up FSx for ONTAP

A managed storage service like FSx for ONTAP should be easy to set up and configure. NetApp Console simplifies this process while at the same time gives administrators access to advanced features that, in contrast to other fully-managed storage services, only Amazon FSx for NetApp ONTAP can support.

In this activity, you create a new FSx for ONTAP file system using NetApp Console, and observe all the options available for creating using recommended best practices.

- On the NetApp Console Home page, expand the left navigation menu using the three horizontal lines at the top, click **Storage > Management**.



Note: From NetApp Console there are two paths you can take to create an FSx for ONTAP file system using NetApp Workload Factory. Workload Factory offers a few options for FSx for ONTAP creation by navigating to **Workloads > Workloads Overview > Create FSx for ONTAP**, or by navigating to **Workloads > Storage > Dashboard > Create file system**, or by navigating to **Workloads > Storage > FSx for ONTAP > Create file**

system. Regardless of which path you take in NetApp Console or NetApp Workload Factory you will have an identical experience.

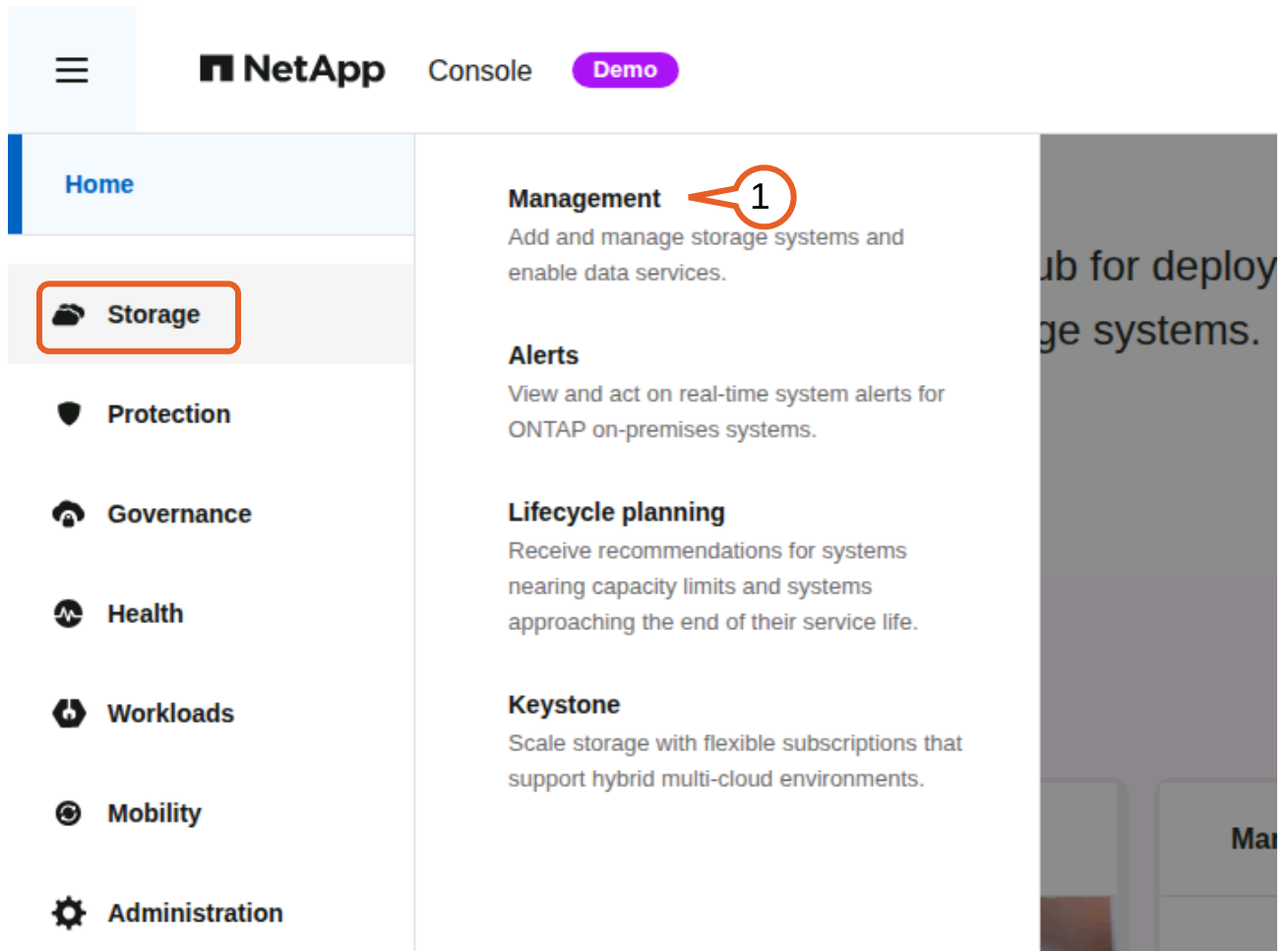


Figure 2-2:

2. In the Systems View, click **Add system**.

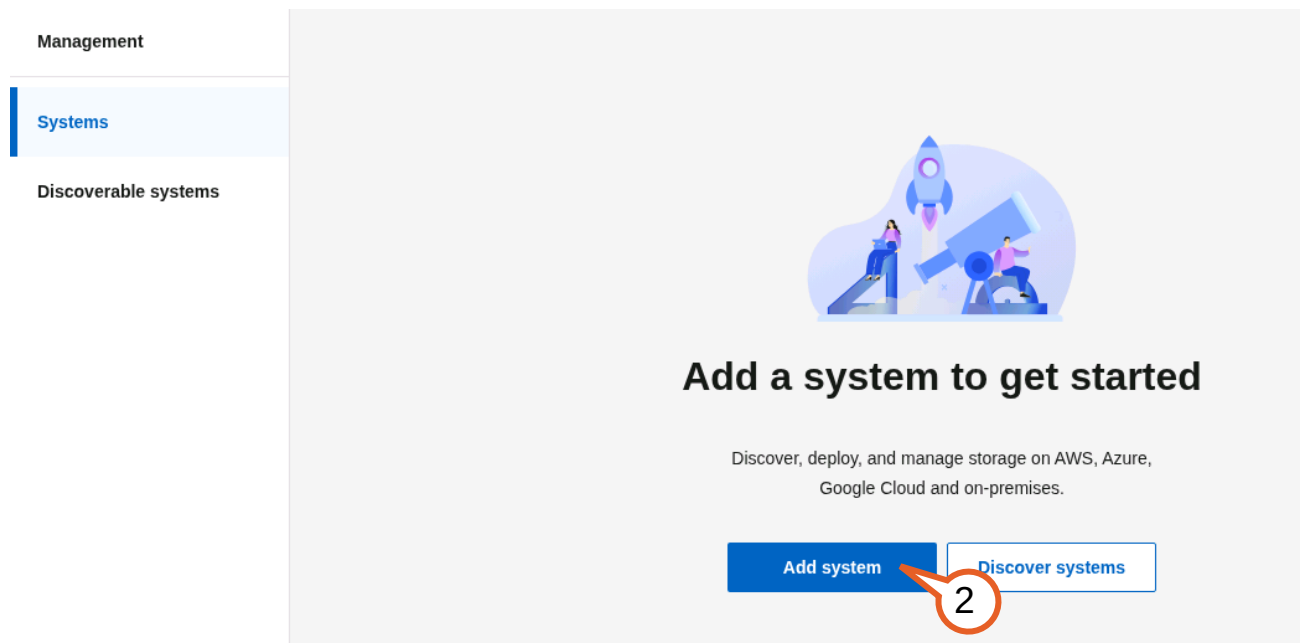


Figure 2-3:

3. Click **Amazon Web Services**.

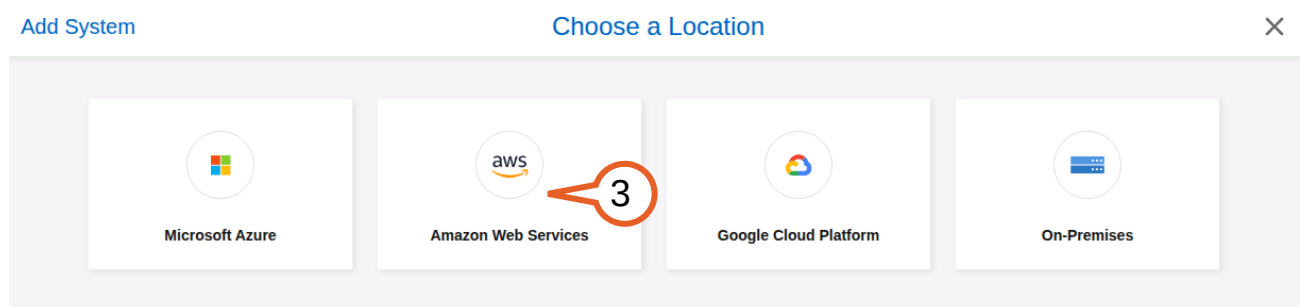


Figure 2-4:

- On the line for Amazon FSx for ONTAP, click **Add new**.

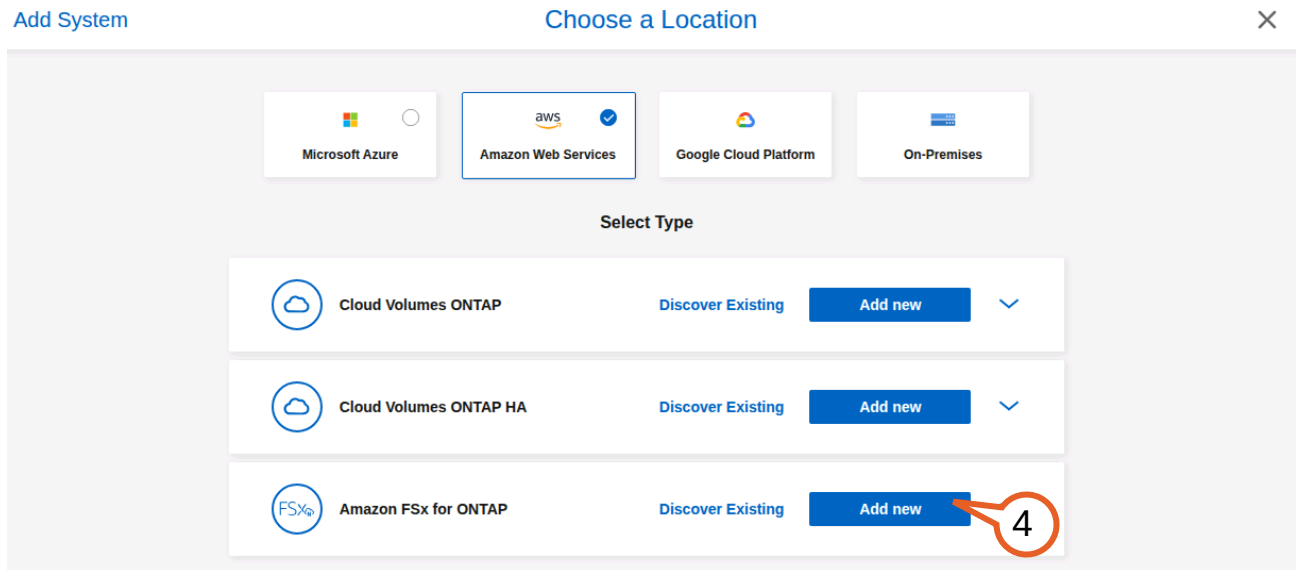


Figure 2-5:

- Before you start take a moment to look at the `</>` **Codebox** at the right and select the **drop-down menu** to show the three options available: API request, CloudFormation, and Terraform. As you go through the workflows, the Codebox automatically adjusts the parameters and gives you an easy way to convert any task into a configuration you can use in your preferred automation tool.



Note: Codebox is an infrastructure as code (IaC) co-pilot that helps you generate code to execute any operations supported by NetApp Workloads. To learn more about how to use Codebox to more easily generate code visit: <https://docs.netapp.com/us-en/workload-setup-admin/codebox-automation.html>.

- Click **Advanced create**.

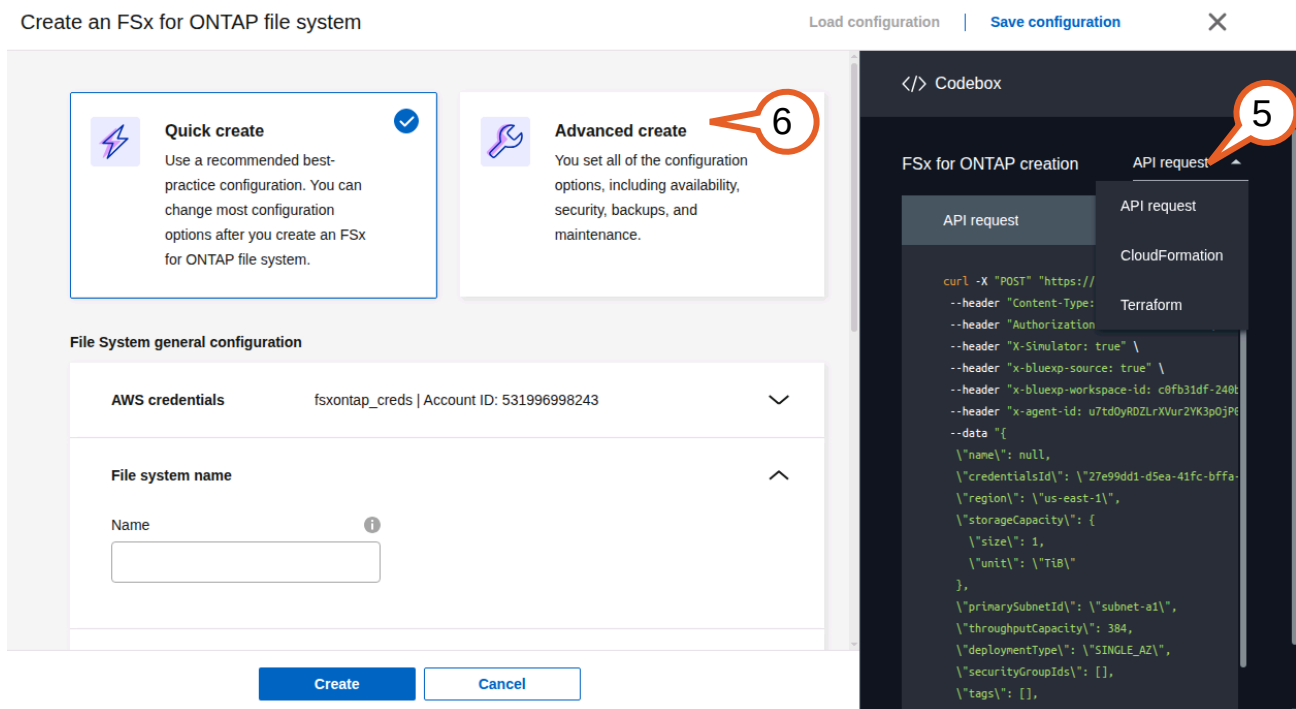


Figure 2-6:

- Click the **downward arrow** next to “File system name” and enter **fsxontap** for the name.



Tip: The arrows in the UI point up when a section is expanded and clicking the up arrow will collapse the section. Downward arrows in the UI indicate the section is currently collapsed and clicking on the downward arrow will expand the selection.

Create an FSx for ONTAP file system

Load configuration | Save configuration X

Quick create
Use a recommended best-practice configuration. You can change most configuration options after you create an FSx for ONTAP file system.

Advanced create ✓
You set all of the configuration options, including availability, security, backups, and maintenance.

File System general configuration

AWS credentials fsxontap_creds | Account ID: 531996998243

File system name

Name
fsxontap 7

Create Cancel

Codebox

FSx for ONTAP creation API request

```
API request
curl -X "POST" "https://staging.api.workloads.r
--header "Content-Type: application/json" \
--header "Authorization: <Bearer Token>" \
--header "X-Simulator: true" \
--header "x-bluexp-source: true" \
--header "x-bluexp-workspace-id: c0fb31df-240b
--header "x-agent-id: u7td0y@DZLrXVur2YK3p0jP6
--data "{
  \"name\": \"fsxontap\",
  \"credentialsId\": \"27e99dd1-d5ea-41fc-bffa-
  \"region\": \"us-east-1\",
  \"storageCapacity\": {
    \"size\": 1,
    \"unit\": \"TiB\"
  },
  \"primarySubnetId\": \"subnet-a1\",
  \"throughputCapacity\": 384,
  \"deploymentType\": \"SINGLE_AZ\",
  \"securityGroupIds\": [],
  \"tags\": []
}
```

Figure 2-7:

- Click the **downward arrow** next to “Deployment type” and select **Multiple Availability Zones**.

Takeaway: FSx for ONTAP file systems are highly available and fail over and back as needed. Clients automatically fail over and back with the file system.

Create an FSx for ONTAP file system Load configuration Save configuration ✕

Region & VPC US East (N. Virginia) | us-east-1 | vpc-1 - 172....

Deployment type ⌵

Select a deployment type:

☒ **Single Availability Zone**
Single-AZ file systems require replicating workloads to a file system in another AZ to safeguard against instance failures and AZ disruptions.

File system generation:

☒ Second-generation

☐ First-generation

8

☐ **Multiple Availability Zones**
Multi-AZ file systems have a standby server in a second Availability Zone (AZ) to protect your data against instance failures and AZ disruptions.

Create

Cancel

</> Codebox

FSx for ONTAP creation API request ⌵

API request 📄

```
curl -X "POST" "https://staging.api.workloads.r
--header "Content-Type: application/json" \
--header "Authorization: <Bearer Token>" \
--header "X-Simulator: true" \
--header "x-bluexp-source: true" \
--header "x-bluexp-workspace-id: c0fb31df-240b
--header "x-agent-id: u7td0yR0ZLrXVur2YK3p0JPt
--data "{
  \"name\": \"fsxontap\",
  \"credentialsId\": \"27e99dd1-d5ea-41fc-bffa-
  \"region\": \"us-east-1\",
  \"storageCapacity\": {
    \"size\": 1,
    \"unit\": \"TiB\"
  },
  \"primarySubnetId\": \"subnet-a1\",
  \"throughputCapacity\": 384,
  \"deploymentType\": \"SINGLE_AZ\",
  \"securityGroupIds\": [],
  \"tags\": [],
  \"state\": \"PENDING\"
}
```

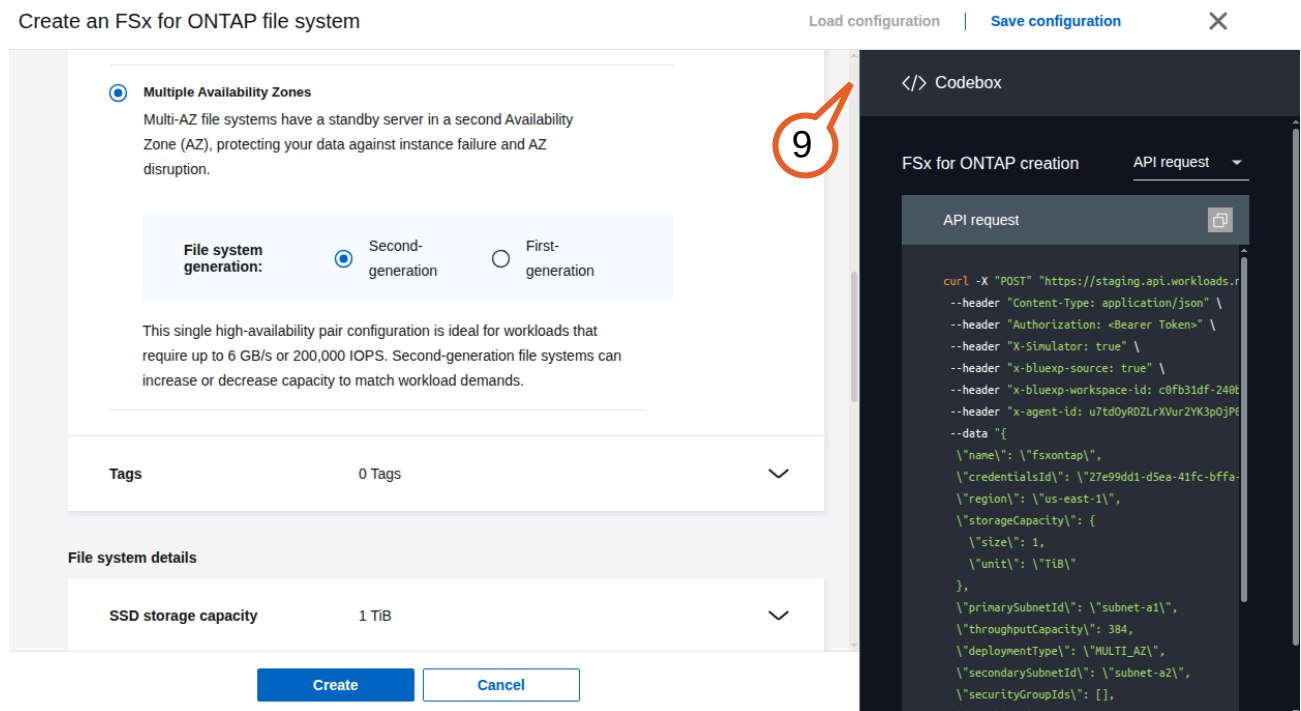
Figure 2-8:

11

Getting Started with FSx for ONTAP
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- After reviewing the Multi-AZ file system details, scroll back to the top under Region & VPC.

 **Note:** If the scroll bar is difficult to see, you can also click in the window and use the up and down arrows on your keyboard.



Create an FSx for ONTAP file system

Load configuration | Save configuration X

Multiple Availability Zones

Multi-AZ file systems have a standby server in a second Availability Zone (AZ), protecting your data against instance failure and AZ disruption.

File system generation: ☒ Second-generation ☐ First-generation

This single high-availability pair configuration is ideal for workloads that require up to 6 GB/s or 200,000 IOPS. Second-generation file systems can increase or decrease capacity to match workload demands.

Tags 0 Tags

File system details

SSD storage capacity 1 TiB

Create Cancel

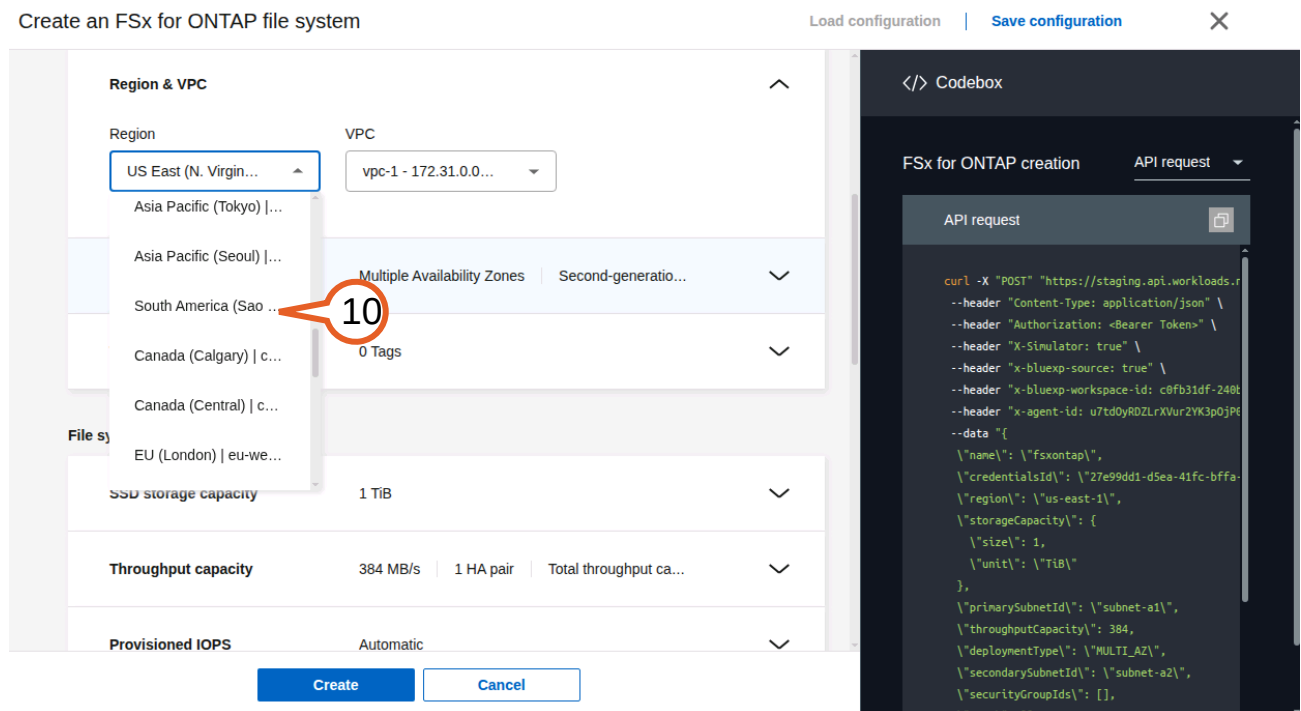
Codebox

FSx for ONTAP creation API request

```
curl -X "POST" "https://staging.api.workloads.r
--header "Content-Type: application/json" \
--header "Authorization: <Bearer Token>" \
--header "X-Simulator: true" \
--header "x-bluelxp-source: true" \
--header "x-bluelxp-workspace-id: c0fb31df-240t
--header "x-agent-id: u7td0yR0ZLrXVur2YK3p0jP4
--data "{
  \"name\": \"fsxontap\",
  \"credentialsId\": \"27e99dd1-d5ea-41fc-bffa-
  \"region\": \"us-east-1\",
  \"storageCapacity\": {
    \"size\": 1,
    \"unit\": \"TiB\"
  },
  \"primarySubnetId\": \"subnet-a1\",
  \"throughputCapacity\": 384,
  \"deploymentType\": \"MULTI_AZ\",
  \"secondarySubnetId\": \"subnet-a2\",
  \"securityGroupIds\": [],
  \"agentId\": \"u7td0yR0ZLrXVur2YK3p0jP4\"
}
```

Figure 2-9:

- Now change the Region in the drop-down menu to **South America (Sao Paulo) | sa-east-1**.



Create an FSx for ONTAP file system

Load configuration | Save configuration X

Region & VPC

Region VPC

US East (N. Virgin... vpc-1 - 172.31.0.0...

Asia Pacific (Tokyo) |... Multiple Availability Zones Second-generation...

Asia Pacific (Seoul) |... 0 Tags

South America (Sao Paulo) | sa-east-1 10

Canada (Calgary) | c... 1 TiB

Canada (Central) | c...

EU (London) | eu-we...

File system details

SSD storage capacity 1 TiB

Throughput capacity 384 MB/s 1 HA pair Total throughput ca...

Provisioned IOPS Automatic

Create Cancel

Codebox

FSx for ONTAP creation API request

```
curl -X "POST" "https://staging.api.workloads.r
--header "Content-Type: application/json" \
--header "Authorization: <Bearer Token>" \
--header "X-Simulator: true" \
--header "x-bluelxp-source: true" \
--header "x-bluelxp-workspace-id: c0fb31df-240t
--header "x-agent-id: u7td0yR0ZLrXVur2YK3p0jP4
--data "{
  \"name\": \"fsxontap\",
  \"credentialsId\": \"27e99dd1-d5ea-41fc-bffa-
  \"region\": \"us-east-1\",
  \"storageCapacity\": {
    \"size\": 1,
    \"unit\": \"TiB\"
  },
  \"primarySubnetId\": \"subnet-a1\",
  \"throughputCapacity\": 384,
  \"deploymentType\": \"MULTI_AZ\",
  \"secondarySubnetId\": \"subnet-a2\",
  \"securityGroupIds\": [],
  \"agentId\": \"u7td0yR0ZLrXVur2YK3p0jP4\"
}
```

Figure 2-10:

11. After changing the region, NetApp Workloads immediately alerts you that the region does not support the option.

The screenshot shows the 'Create an FSx for ONTAP file system' console. The 'Region & VPC' section is expanded, showing 'Region' set to 'South America (S...)' and 'VPC' set to 'vpc-1 - 172.31.0.0...'. A warning message states: 'This region doesn't support second-generation FSx for ONTAP file systems.' A red circle with the number '11' highlights this message. The 'Deployment type' is set to 'Multiple Availability Zones' and 'First-generation'. The 'Tags' section shows '0 Tags'. The 'File system details' section shows 'SSD storage capacity' as '1 TiB' and 'Throughput capacity' as '128 MB/s', '1 HA pair', and 'Total throughput ca...'. The 'Create' and 'Cancel' buttons are at the bottom. On the right, a 'Codebox' shows the 'API request' for 'FSx for ONTAP creation'.

Figure 2-11:

12. Scroll down and click the **downward arrow** next to “ONTAP credentials” and enter **Netapp1!** for the password, and enter the same a second time to confirm the password.
13. Take note that after entering in the password twice, the Storage VM credentials are automatically entered, but you do have the option in the Advanced Create to manually change them..

The screenshot shows the 'Create an FSx for ONTAP file system' console. The 'ONTAP credentials' section is expanded, showing 'User name' as 'fsxadmin'. The 'Password' and 'Confirm password' fields are both filled with '*****'. A red circle with the number '12' highlights the 'Confirm password' field. The 'Storage VM credentials' section shows 'vsadmin' and 'Same password entered for the FS...'. A red circle with the number '13' highlights this section. The 'SMB/CIFS setup' section shows 'Not configured'. The 'Network & security' section is partially visible. The 'Create' and 'Cancel' buttons are at the bottom. On the right, a 'Codebox' shows the 'API request' for 'FSx for ONTAP creation'.

Figure 2-12:

14. Scroll down to the **FSx for ONTAP backup** section and expand.



Note: FSx for ONTAP backup is service provided by AWS that works for all AWS FSx offerings, but when using FSx for ONTAP you are able to use ONTAP features. This will be discussed further in a later section of this guide.

Create an FSx for ONTAP file system Load configuration | Save configuration ✕

FSx for ONTAP backup

Daily automatic backups

Amazon FSx for ONTAP can protect your data through daily backups.

Automatic backup retention period

Choose the number of days (between 1 and 90 days) that Amazon FSx for ONTAP should retain automatic backups for this file system.

Number of days

Daily automatic backup window

☒ No preference ☐ Select start time for 30-minute weekly backup window

A random backup window will be selected.

Create Cancel

Codebox

FSx for ONTAP creation API request

```
API request
curl -X "POST" "https://staging.api.workloads.r
--header "Content-Type: application/json" \
--header "Authorization: <Bearer Token>" \
--header "X-Simulator: true" \
--header "x-bluexp-source: true" \
--header "x-bluexp-workspace-id: c0fb31df-248t
--header "x-agent-id: u7td0yR0ZLrXVur2YK3p0jPt
--data "{
  \"name\": \"fsxontap\",
  \"credentialsId\": \"27e99dd1-d5ea-41fc-bffa-
  \"region\": \"sa-east-1\",
  \"storageCapacity\": {
    \"size\": 1,
    \"unit\": \"TiB\"
  },
  \"primarySubnetId\": \"subnet-a1\",
  \"throughputCapacity\": 128,
  \"fsxAdminPassword\": \"*****\",
  \"deploymentType\": \"MULTI_AZ\",
  \"secondarySubnetId\": \"subnet-a2\",
  \"secondarySubnetId3\": \"\"
}
```

Figure 2-13:

15. Scroll back to the top of the page and switch to **Quick create**.

16. Click the **downward arrow** next to “File system name” and enter **fsxontap1** for the name.

Create an FSx for ONTAP file system Load configuration | Save configuration ✕

Quick create 15

Use a recommended best-practice configuration. You can change most configuration options after you create an FSx for ONTAP file system.

Advanced create

You set all of the configuration options, including availability, security, backups, and maintenance.

File System general configuration

AWS credentials fsxontap_creds | Account ID: 531996998243

File system name

Name 16

fsxontap1

Create Cancel

Codebox

FSx for ONTAP creation API request

API request

```
curl -X "POST" "https://staging.api.workloads.r
--header "Content-Type: application/json" \
--header "Authorization: <Bearer Token>" \
--header "X-Simulator: true" \
--header "x-bluexp-source: true" \
--header "x-bluexp-workspace-id: c0fb31df-248b
--header "x-agent-id: u7td0yR0ZLrXVur2YK3p0jP6
--data "{
  \"name\": \"fsxontap1\",
  \"credentialsId\": \"27e99dd1-d5ea-41fc-bffa-
  \"region\": \"us-east-1\",
  \"storageCapacity\": {
    \"size\": 1,
    \"unit\": \"TiB\"
  },
  \"primarySubnetId\": \"subnet-a1\",
  \"throughputCapacity\": 384,
  \"deploymentType\": \"SINGLE_AZ\",
  \"securityGroupIds\": [],
  \"tags\": [],
  \"securityGroupIds\": [],
  \"tags\": [],
  \"securityGroupIds\": [],
  \"tags\": []
}
```

Figure 2-14:



Note: If you do not see AWS credentials that start with “fsxontap” and instead see “DemoDefaultCredential”, you may need to perform an additional step before creating your first FSx for ONTAP file system. Start by expanding the AWS credentials section and click on the link that takes you to the **Credentials page**. Follow the wizard to add credentials with the storage permissions policy. Prefix the name of the new credentials you add with **fsxontap...** Then return to the Quick create wizard to continue with the steps below.

- Before continuing you need to set a password for the fsxadmin account. Scroll down to ONTAP credentials.

Create an FSx for ONTAP file system Load configuration | Save configuration ✕

File system name	fsxontap1	▼
Region & VPC	US East (N. Virginia) us-east-1 vpc-1 - 172.31.0.0/...	▼
Deployment type	Single Availability Zone	▼
Tags	0 Tags	▼

File system details

SSD storage capacity	1 TiB	▼
ONTAP credentials	fsxadmin Password not set	▼
SMB/CIFS setup	Not configured	▼

Create Cancel

</> Codebox

FSx for ONTAP creation API request ▼

API request

```
curl -X "POST" "https://staging.api.workloads.r
--header "Content-Type: application/json" \
--header "Authorization: <Bearer Token>" \
--header "X-Simulator: true" \
--header "x-bluexp-source: true" \
--header "x-bluexp-workspace-id: c0fb31df-248b
--header "x-agent-id: u7td0yR0ZLrXVur2YK3p0jP6
--data "{
  \"name\": \"fsxontap1\",
  \"credentialsId\": \"27e99dd1-d5ea-41fc-bffa-
  \"region\": \"us-east-1\",
  \"storageCapacity\": {
    \"size\": 1,
    \"unit\": \"TiB\"
  },
  \"primarySubnetId\": \"subnet-a1\",
  \"throughputCapacity\": 384,
  \"deploymentType\": \"SINGLE_AZ\",
  \"securityGroupIds\": [],
  \"tags\": [],
```

Figure 2-15:

- Enter **Netapp1!** for the password, and enter the same a second time to confirm the password.



Note: If you recall from the *Advanced create* you were given the option to set the Storage VM credentials, but that option is not available in *Quick create* to simplify the creation.

19. At the top right click **Save configuration**.

Create an FSx for ONTAP file system

Load configuration | **Save configuration** X

ONTAP credentials

Entering a password for the FSx admin user is optional. The administrative user for this FSx for ONTAP file system can be changed later.

User name

fsxadmin

Password Confirm password

18

SMB/CIFS setup Not configured

Summary

Default configurations View the default configuration created by the system

Create Cancel

Codebox

FSx for ONTAP creation API request

```
curl -X "POST" "https://staging.api.workloads.r
--header "Content-Type: application/json" \
--header "Authorization: <Bearer Token>" \
--header "X-Simulator: true" \
--header "x-bluexp-source: true" \
--header "x-bluexp-workspace-id: c0fb31df-248b
--header "x-agent-id: u7td0yR0ZLrXVur2YK3p0Jp6
--data "{
  \"name\": \"fsxontap1\",
  \"credentialsId\": \"27e99dd1-d5ea-41fc-bffa-
  \"region\": \"us-east-1\",
  \"storageCapacity\": {
    \"size\": 1,
    \"unit\": \"TiB\"
  },
  \"primarySubnetId\": \"subnet-a1\",
  \"throughputCapacity\": 384,
  \"fsxAdminPassword\": \"*****\",
  \"deploymentType\": \"SINGLE_AZ\",
  \"securityGroupIds\": [],
  \"agentId\": \"\"
}
```

Figure 2-16:

20. From here you can save your settings and load them again at a future time. Click **Cancel**.

Save FSx for ONTAP configuration

You can save this FSx for ONTAP configuration and load the configuration later for a future FSx for ONTAP deployment.

Configuration name

Save Cancel 20

Figure 2-17:

21. Click **Create**.

Create an FSx for ONTAP file system

Load configuration | Save configuration X

ONTAP credentials

Entering a password for the FSx admin user is optional. The administrative user for this FSx for ONTAP file system can be changed later.

User name
fsxadmin

Password Confirm password

SMB/CIFS setup Not configured

Summary
Default configurations View the default configuration created by the system

Create Cancel

Codebox

FSx for ONTAP creation API request

```
curl -X "POST" "https://staging.api.workloads.r
--header "Content-Type: application/json" \
--header "Authorization: <Bearer Token>" \
--header "X-Simulator: true" \
--header "x-bluexp-source: true" \
--header "x-bluexp-workspace-id: c0fb31df-248b
--header "x-agent-id: u7td0yR0ZLrXVur2YK3p0JPt
--data "{
  \"name\": \"fsxontap1\",
  \"credentialsId\": \"27e99dd1-d5ea-41fc-bffa-
  \"region\": \"us-east-1\",
  \"storageCapacity\": {
    \"size\": 1,
    \"unit\": \"TiB\"
  },
  \"primarySubnetId\": \"subnet-a1\",
  \"throughputCapacity\": 384,
  \"fsxAdminPassword\": \"*****\",
  \"deploymentType\": \"SINGLE_AZ\",
  \"securityGroupIds\": [],
  \"agentId\": \"f3
```

Figure 2-18:

22. The status of the fsxontap1 instance will appear as “Initializing” for just a moment. Once it appears as shown here, the environment is ready for you to continue.

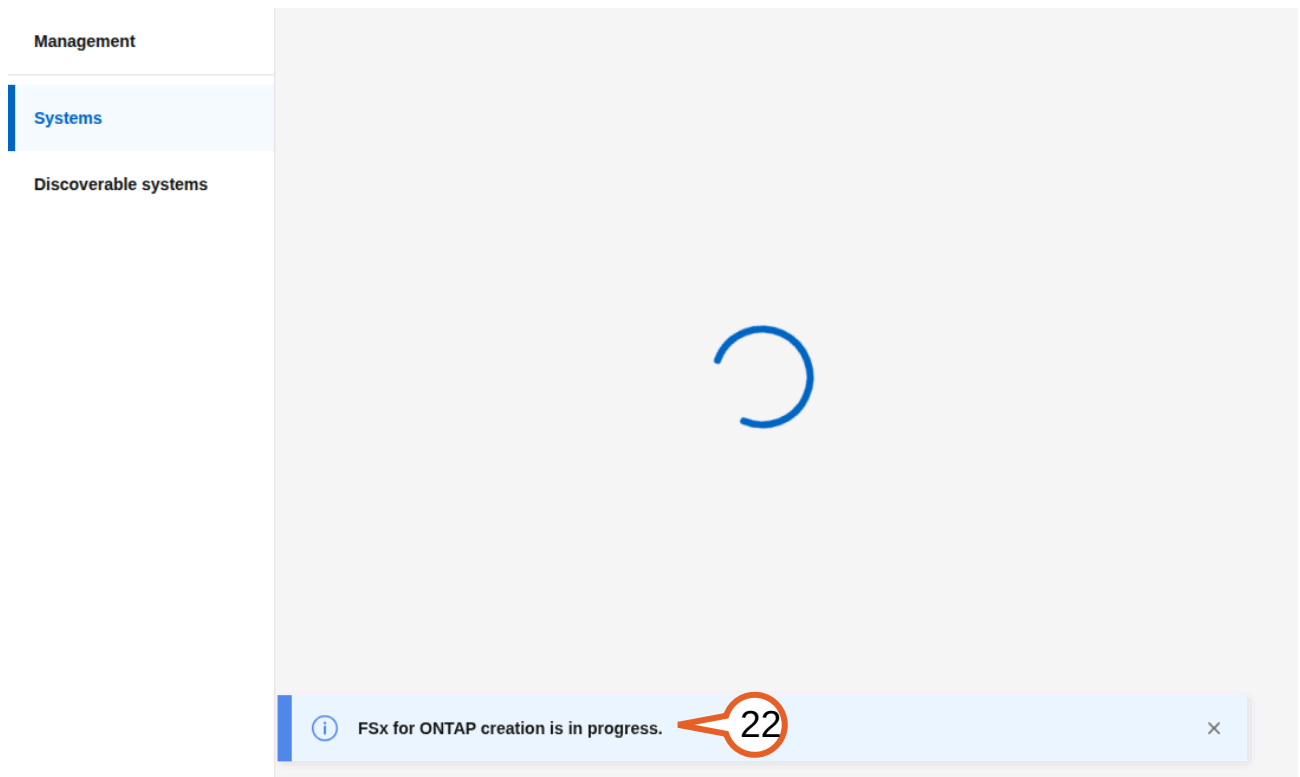


Figure 2-19:

23. When you are in Systems Management View, you will see the file system appear in a cloud icon when complete.
24. Now begin creating a second file system by clicking + Add.

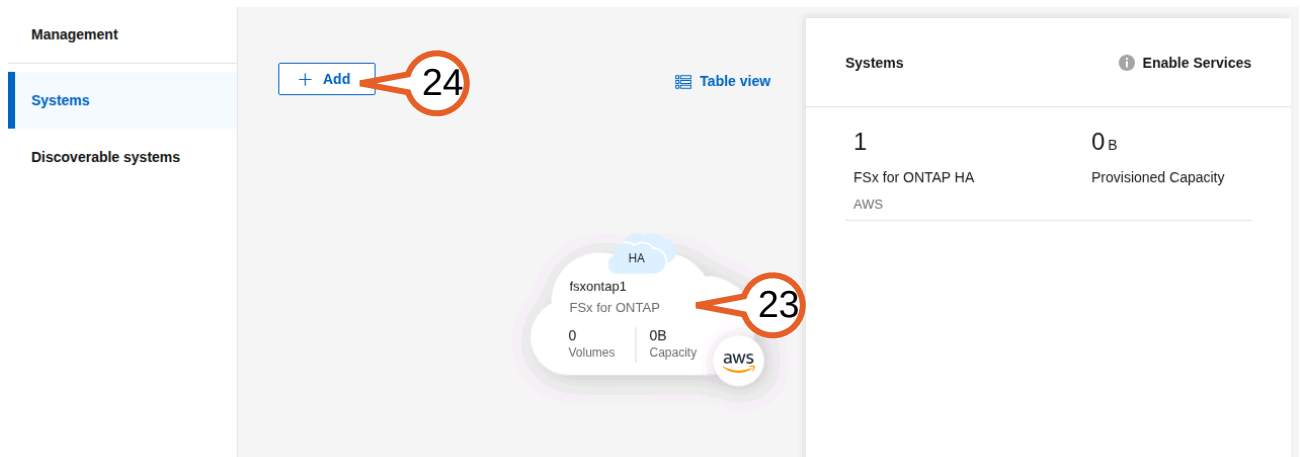


Figure 2-20:

25. Click **Amazon Web Services**.

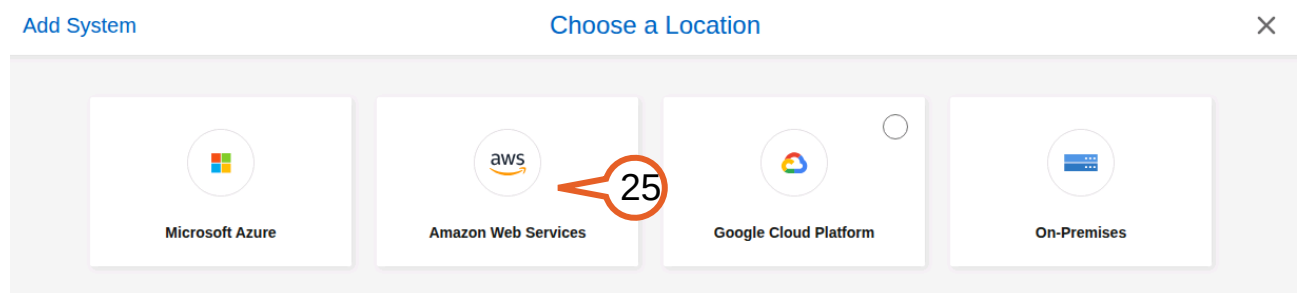


Figure 2-21:

26. On the line for Amazon FSx for ONTAP, click **Add new**.

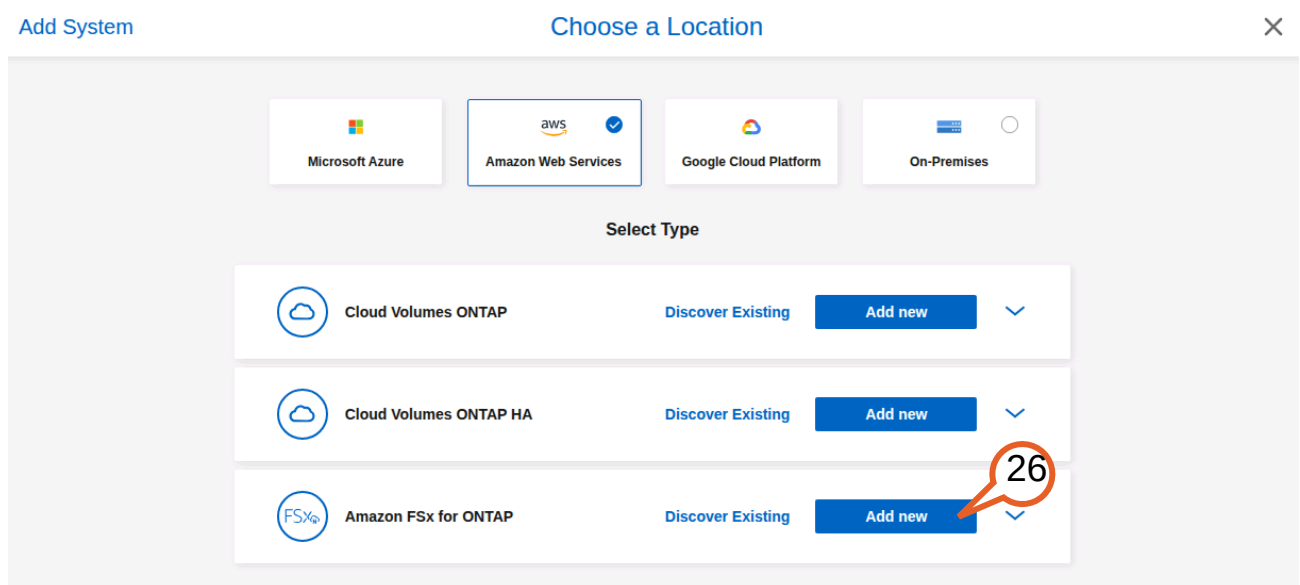


Figure 2-22:

27. Use **Quick create**.
28. Click the **downward arrow** next to “File system name” and enter **fsxontap2** for the name.

Create an FSx for ONTAP file system Load configuration | Save configuration ✕

Quick create 27

Use a recommended best-practice configuration. You can change most configuration options after you create an FSx for ONTAP file system.

Advanced create

You set all of the configuration options, including availability, security, backups, and maintenance.

File System general configuration

AWS credentials fsxontap_creds | Account ID: 531996998243

File system name

Name

fsxontap2 28

Create Cancel

Codebox

FSx for ONTAP creation API request

API request

```
curl -X "POST" "https://staging.api.workloads.r
--header "Content-Type: application/json" \
--header "Authorization: <Bearer Token>" \
--header "X-Simulator: true" \
--header "x-bluelxp-source: true" \
--header "x-bluelxp-workspace-id: c0fb31df-248t
--header "x-agent-id: u7td0yR0ZLrKVur2YK3p0jP6
--data "{
  \"name\": \"fsxontap2\",
  \"credentialsId\": \"27e99dd1-d5ea-41fc-bffa-
  \"region\": \"us-east-1\",
  \"storageCapacity\": {
    \"size\": 1,
    \"unit\": \"TiB\"
  },
  \"primarySubnetId\": \"subnet-a1\",
  \"throughputCapacity\": 384,
  \"deploymentType\": \"SINGLE_AZ\",
  \"securityGroupIds\": [],
  \"tags\": [],
  \"fsxAdminPassword\": \"*****\",
  \"deploymentType\": \"SINGLE_AZ\",
  \"securityGroupIds\": [],
```

Figure 2-23:

29. Enter **Netapp1!** for the password, and enter the same a second time to confirm the password.
30. Click **Create**.

Create an FSx for ONTAP file system Load configuration | Save configuration ✕

Deployment type Single Availability Zone

Tags 0 Tags

File system details

SSD storage capacity 1 TiB

ONTAP credentials

Entering a password for the FSx admin user is optional. The administrative user for this FSx for ONTAP file system can be changed later.

User name

fsxadmin

Password

***** 30

Confirm password

***** 29

Create Cancel

Codebox

FSx for ONTAP creation API request

API request

```
curl -X "POST" "https://staging.api.workloads.r
--header "Content-Type: application/json" \
--header "Authorization: <Bearer Token>" \
--header "X-Simulator: true" \
--header "x-bluelxp-source: true" \
--header "x-bluelxp-workspace-id: c0fb31df-248t
--header "x-agent-id: u7td0yR0ZLrKVur2YK3p0jP6
--data "{
  \"name\": \"fsxontap2\",
  \"credentialsId\": \"27e99dd1-d5ea-41fc-bffa-
  \"region\": \"us-east-1\",
  \"storageCapacity\": {
    \"size\": 1,
    \"unit\": \"TiB\"
  },
  \"primarySubnetId\": \"subnet-a1\",
  \"throughputCapacity\": 384,
  \"fsxAdminPassword\": \"*****\",
  \"deploymentType\": \"SINGLE_AZ\",
  \"securityGroupIds\": [],
```

Figure 2-24:

31. Once the second file system is ready you will see two file systems in the System Management View with volumes yet to be created.

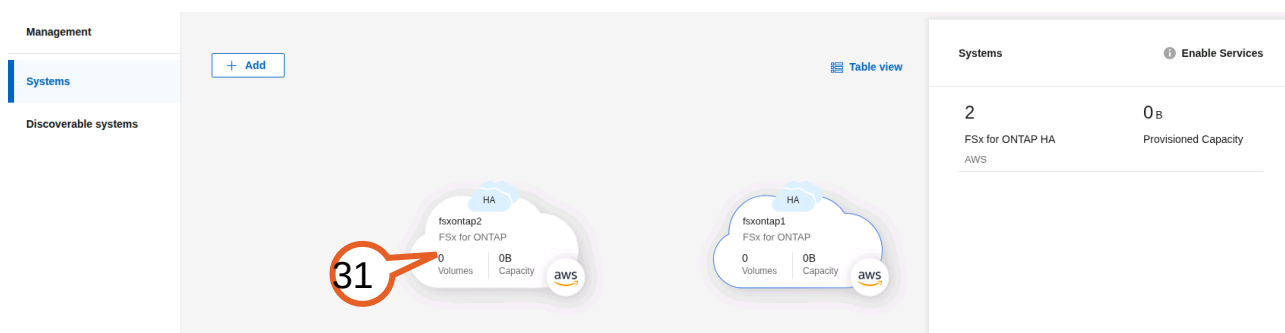


Figure 2-25:

Takeaway: Because of its integration with NetApp Workloads, Amazon FSx for NetApp ONTAP is easily deployed and managed alongside other NetApp products and services in a single interface view in the console. In contrast to what is available with the AWS Console, NetApp Console enables users to leverage advanced data services like backup and recovery, or replication through SnapMirror, by placing it in context with other storage environments.

2.3 Provision and Manage Amazon FSx for NetApp ONTAP Volumes

With the working environment created, your next step is to provision and manage the volumes.

Despite the breadth of features and versatility that it provides, getting started with your new Amazon FSx for NetApp ONTAP file system is as easy as managing any other ONTAP system. Volumes are easy to provision with each of the available protocols (NFS, CIFS, and iSCSI), along with advanced features like storage tiering. Once a volume is provisioned, NetApp Console exposes controls that simplify data management and take advantage of the powerful ONTAP features that the service offers.

In this section of the lab, you use the Console to manage FSx for ONTAP storage and perform the following activities:

Activities
Provision a Volume with Storage Efficiency and Tiering
Explore Data Management Features

2.3.1 Provision a Volume with Storage Efficiency and Tiering

When you deploy Amazon FSx for NetApp ONTAP, a storage virtual machine (SVM) is created automatically. While an Amazon FSx for NetApp ONTAP file system can support multiple SVMs, this lab only uses one. Each SVM can contain one or more volumes, which are the distinct logical data containers that make up your storage. When you create a volume, that volume draws from the total capacity that you configured for the Amazon FSx for NetApp ONTAP service. Volumes are thin provisioned, meaning that they consume storage capacity only for the data stored in them.

In this activity, you use the NetApp Workloads console to create a new FSx for ONTAP volume using the NFS protocol. As part of this process, the volume will have both storage efficiency and storage tiering enabled at its creation. These two features are not available on any other fully-managed storage service from AWS, and NetApp Workloads makes it simple to enable them for a volume.

1. You should still be on the Systems tab within NetApp Console.

Click the **fsxontap1** working environment.

2. In the menu that appears on the right, click **Enter System**.

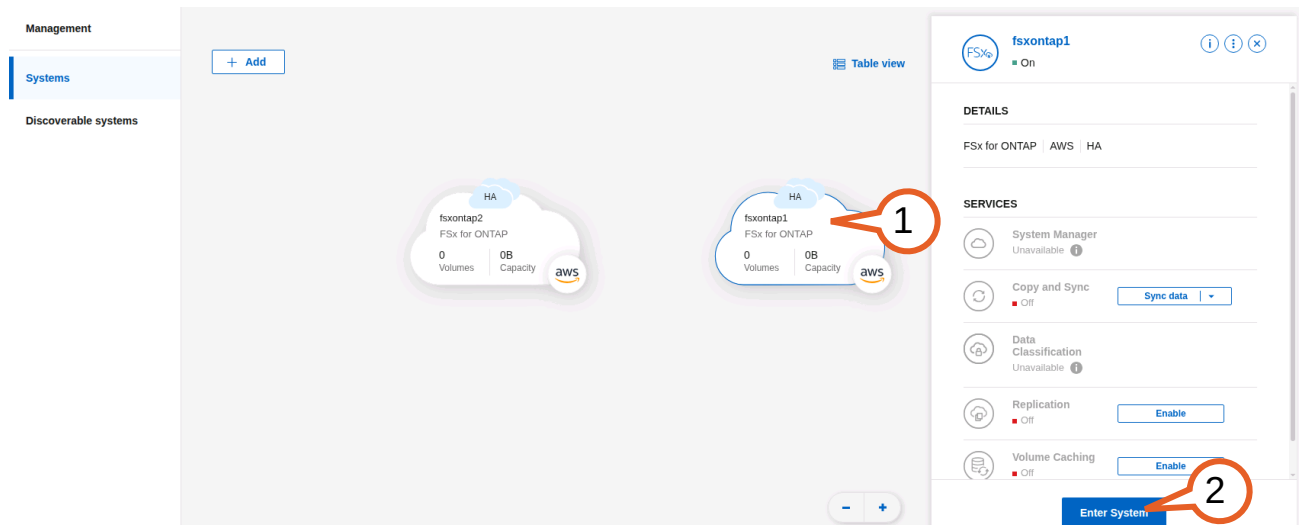


Figure 2-26:

3. You are in an overview of the “fsxontap1” file system. Here you can see information about this file system. Feel free to look through the elements on this tab.
4. To start creating and managing volumes, click + **Create Volume**.

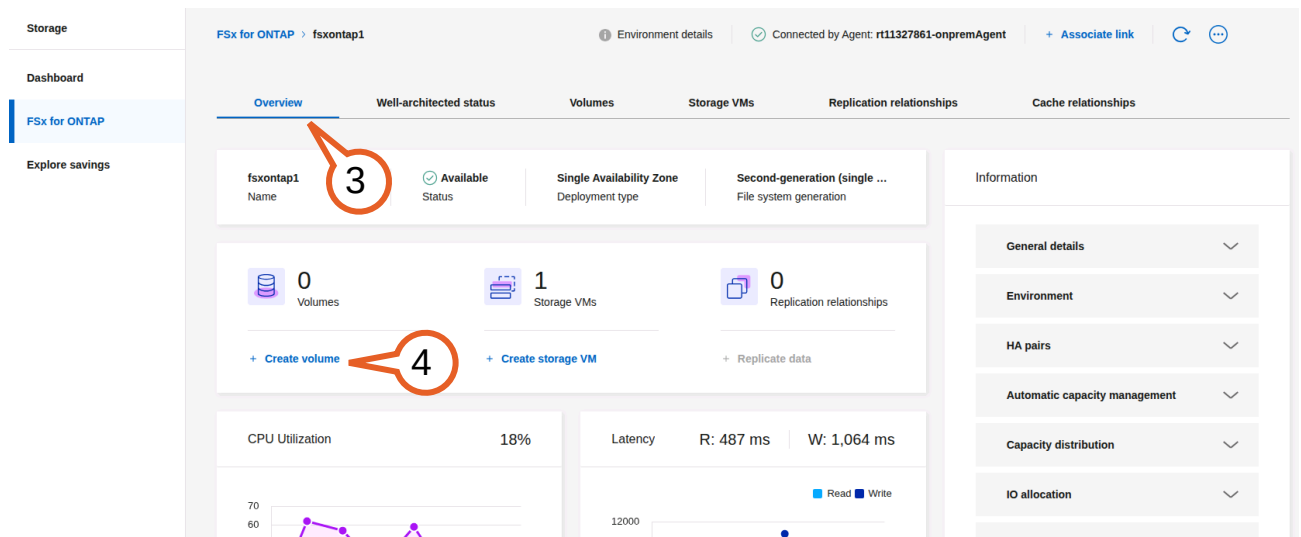


Figure 2-27:

5. Under the General details section named Volume name, enter **fsxvol1**.

6. Scroll down to the Volume size section.



Tip: This is another good place in the UI to pause and notice the JSON details provided in the `</>` Codebox pane. Try changing the drop-down menu to **Terraform** and see how the Codebox dynamically changes as you go through the options.

The screenshot shows the 'Create volume' page in the AWS console. The 'General details' section is expanded, showing the following fields:

- Volume name: fsxvol1 (highlighted with a red circle and the number 5)
- Storage VM name: fsxontap1
- Volume style: FlexVol
- Volume size: Action required (highlighted with a red circle and the number 6)

The Codebox pane on the right shows the API request for volume creation, which is a POST request to the `https://staging.api.workloads.netapp.com/` endpoint. The request body is a JSON object containing the volume details.

Figure 2-28:

7. Click the **downward arrow** next to Volume size and enter **100** for the size and select **GiB** for the unit.
8. Scroll down to the Access section.

The screenshot shows the 'Create volume' page in the AWS console. The 'Volume size' section is expanded, showing the following fields:

- Size: 100 (highlighted with a red circle and the number 7)
- Unit: GiB

The 'Access' section is highlighted with a red circle and the number 8. The 'Access type' field is set to 'File access'. The 'NFS export policy' field is set to 'Existing export policy | default'. The 'SMB/CIFS share' field is set to 'Not configured'.

The Codebox pane on the right shows the API request for volume creation, which is a POST request to the `https://staging.api.workloads.netapp.com/` endpoint. The request body is a JSON object containing the volume details.

Figure 2-29:

- Click the **downward arrow** next to Access type and leave the default selection for the NFS configuration to use the Existing export policy.



Note: FSx for ONTAP supports multi-protocol access to data using CIFS, NFSv3, NFSv4.x, SMB v2.0-v3.1.1 (CIFS), and iSCSI protocols. In this lab, you will only create an NFS volume although SMB/CIFS volume creation is a similar process.

The screenshot shows the 'Create volume' page in the AWS Management Console for FSx for ONTAP. The 'Access type' section is highlighted with a red box and a callout bubble containing the number 9. The 'File access' option is selected. Below this, there are dropdown menus for 'NFS export policy' (set to 'Existing export policy'), 'SMB/CIFS share' (set to 'Not configured'), and 'Security style' (set to 'UNIX'). At the bottom are 'Create' and 'Cancel' buttons. On the right, a 'Codebox' is visible showing an API request for volume creation.

Figure 2-30:

- Scroll down to the Efficiency and protection section to see that Storage efficiency is enabled. FSx for ONTAP is able to take advantage of storage efficiency features of ONTAP, including compression, deduplication, and compaction. Deduplication eliminates duplicate data blocks. Data compression compresses the data blocks to reduce the amount of physical storage that is required. Data compaction stores more data in less space to increase storage efficiency.

Takeaway: Depending on the type of data stored, your storage utilization can be a small fraction of what it would be without taking advantage of these storage efficiency technologies. This can result in significant cost savings.

- Notice the “Tiering policy” is set, but leave the **Balanced (Auto)** setting selected.

Takeaway: Tiering allows you to reduce storage costs by automatically transitioning data between primary storage and lower-cost object storage based on your data access patterns. With the Auto policy that you have selected, FSx for ONTAP will automatically move lesser used data blocks in the active file system as well as snapshot copies to the lower cost storage tier. When access patterns change, blocks can seamlessly move back to the more performant storage. This process, as well as the tier in which data resides, is invisible to the end-user or application.

For additional information on tiering, such as how to make use of the other tiering policies, see the [References section](#).

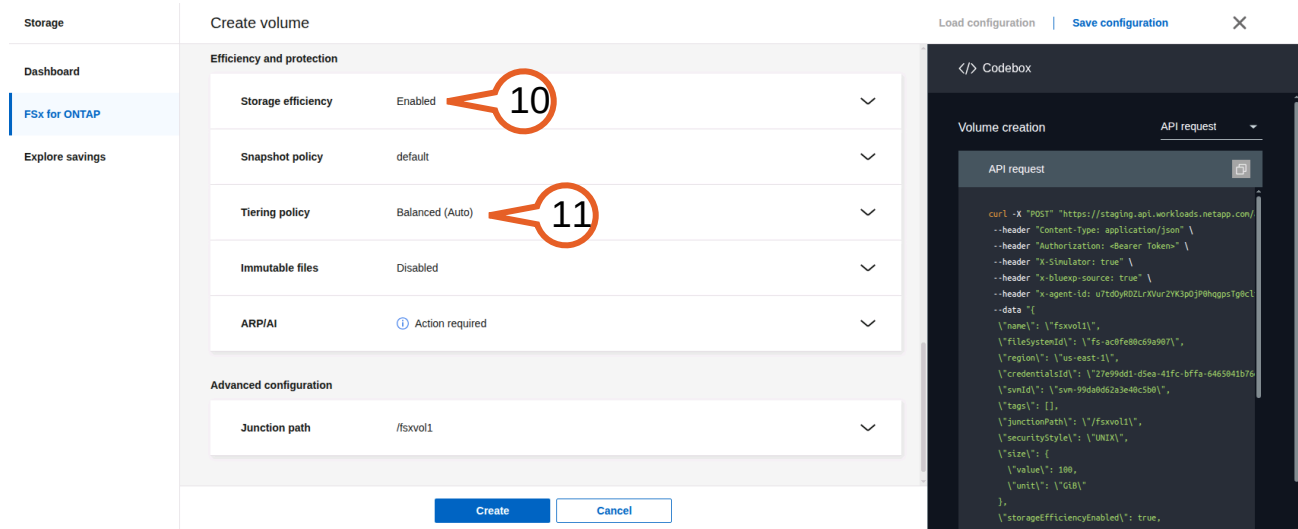


Figure 2-31:

12. Scroll down to the ARPI/AI section. The default allows you to take advantage of NetApp Autonomous Ransomware Protection with AI within ONTAP.
13. Select the **checkbox** to accept the statement to proceed.
14. Click **Create**.

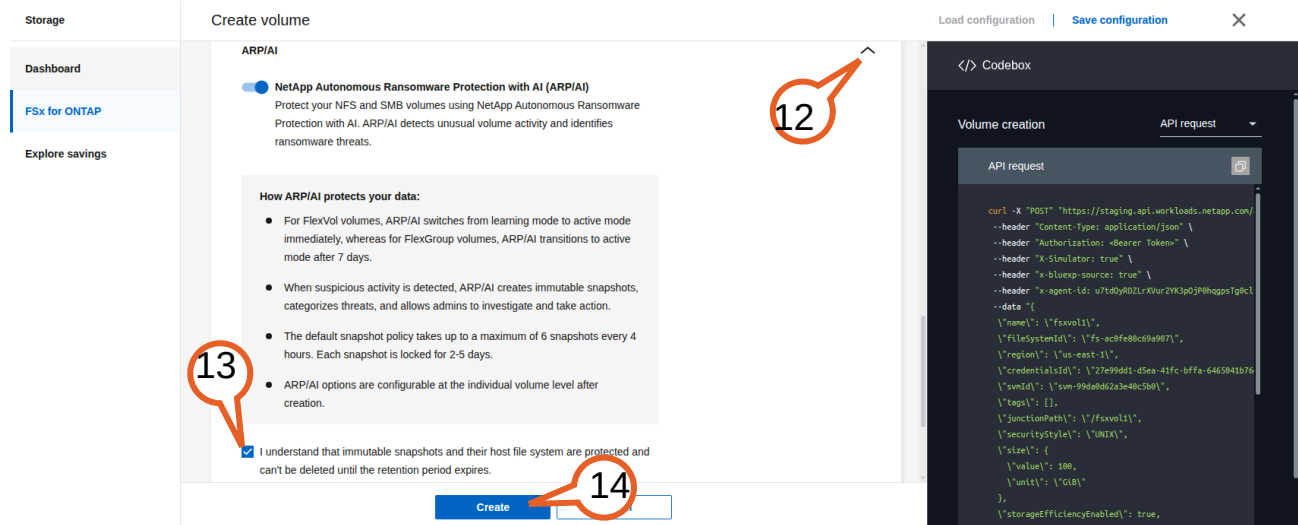


Figure 2-32:

15. Note the location of the refresh button, as you may need it after the next step. Scroll to the bottom of this page.

16. The new volume is added to the FSx for ONTAP working environment. Wait for the state of this new volume to change from “CREATING” to “CREATED,” then continue to the next step.



Important: You might have to click the Refresh icon in the NetApp Workloads console after about 10 seconds to see the updated state shown below.



Tip: You can also create FSx for ONTAP volumes using the FSx for ONTAP console, the AWS CLI, NetApp ONTAP CLI, ONTAP REST APIs, and FSx for ONTAP APIs.

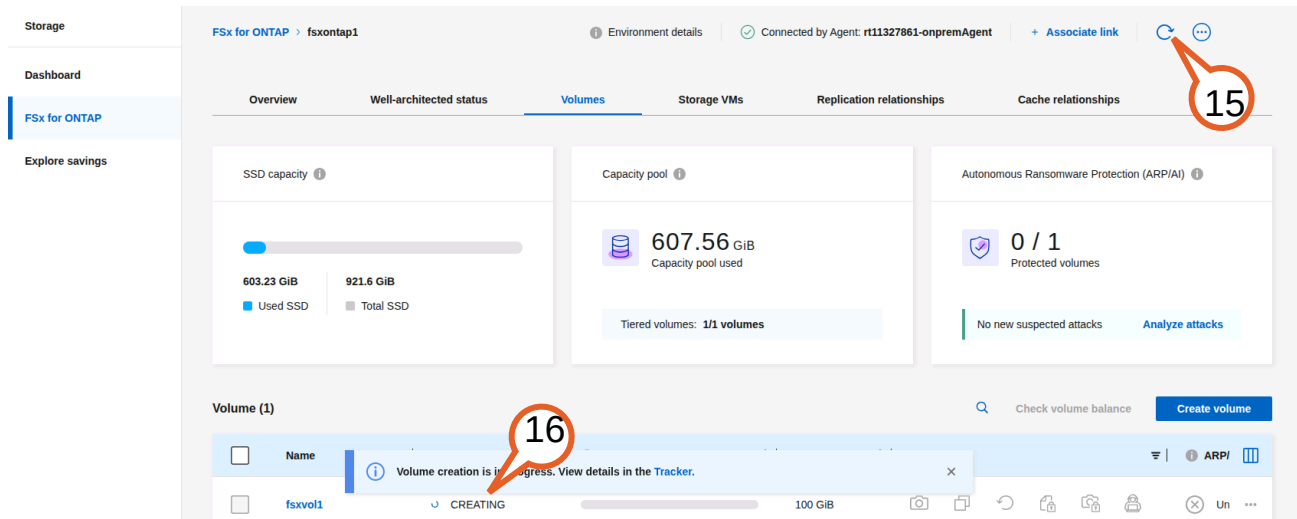


Figure 2-33:

17. Once the volume moves to the status “CREATED” you can hover over the left curving arrow in blue to see that it is also protected because you enabled the FSx for ONTAP backup at the file system level.

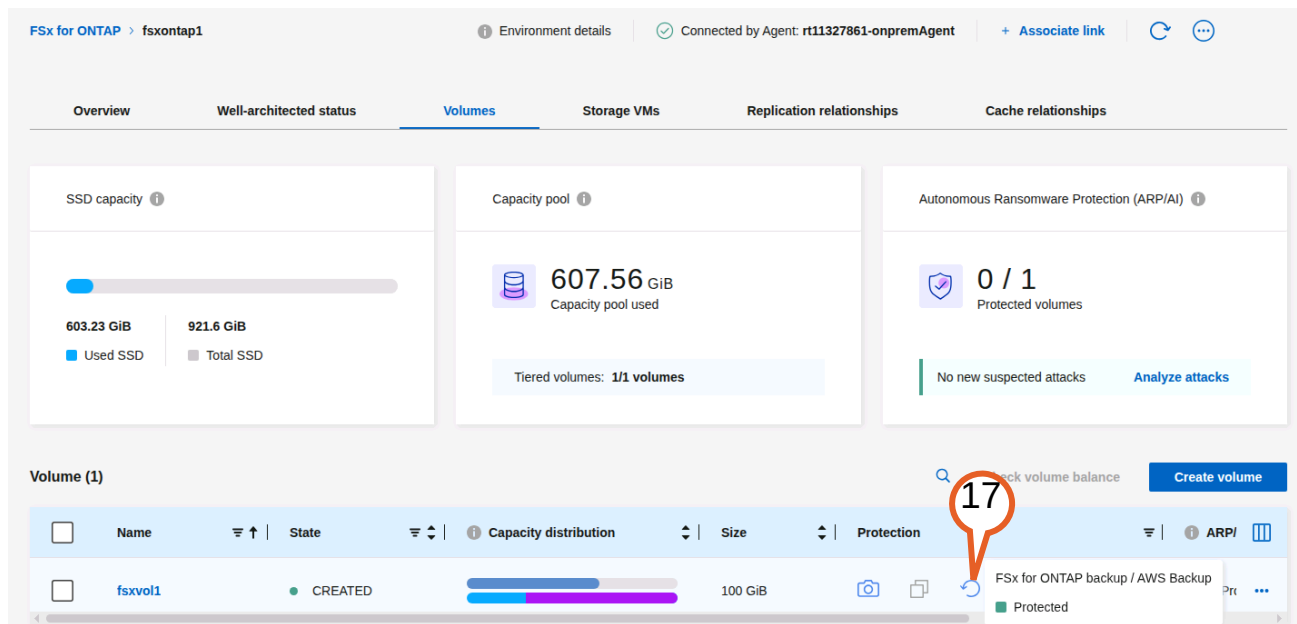


Figure 2-34:

18. Now that the volume has been created, use the following steps to view the volume details. Click the **three horizontal dots** > **Basic actions** > **View volume details** from the menu.

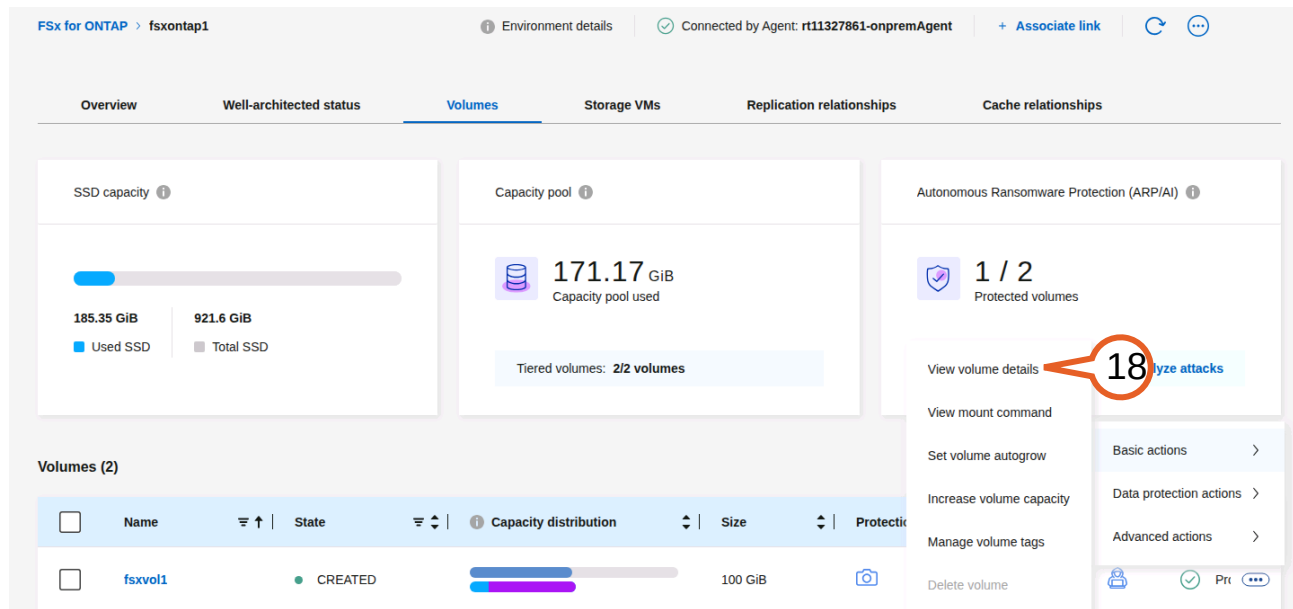


Figure 2-35:

19. Click **Close**.

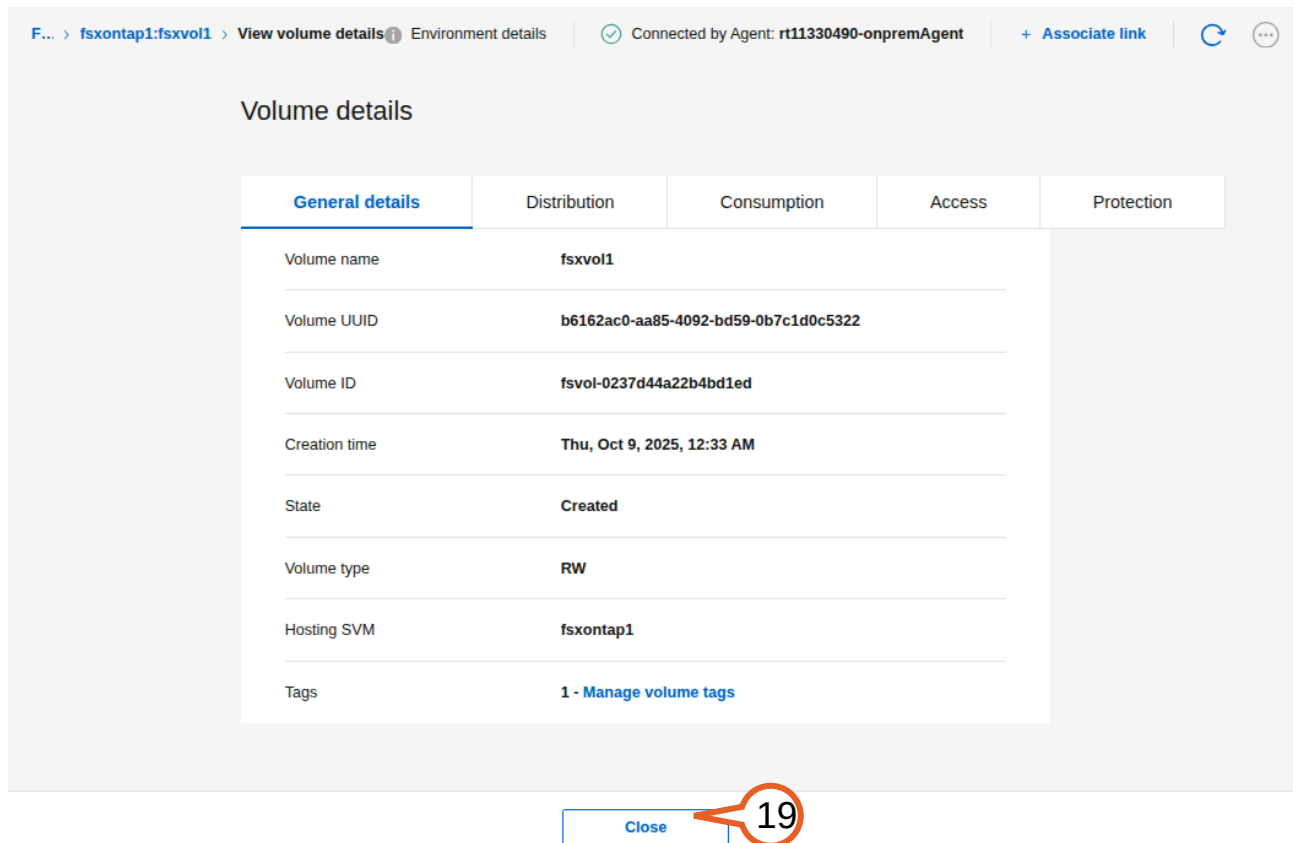


Figure 2-36:

20. Click the **three horizontal dots** > **Basic actions** > **View mount command** from the menu.

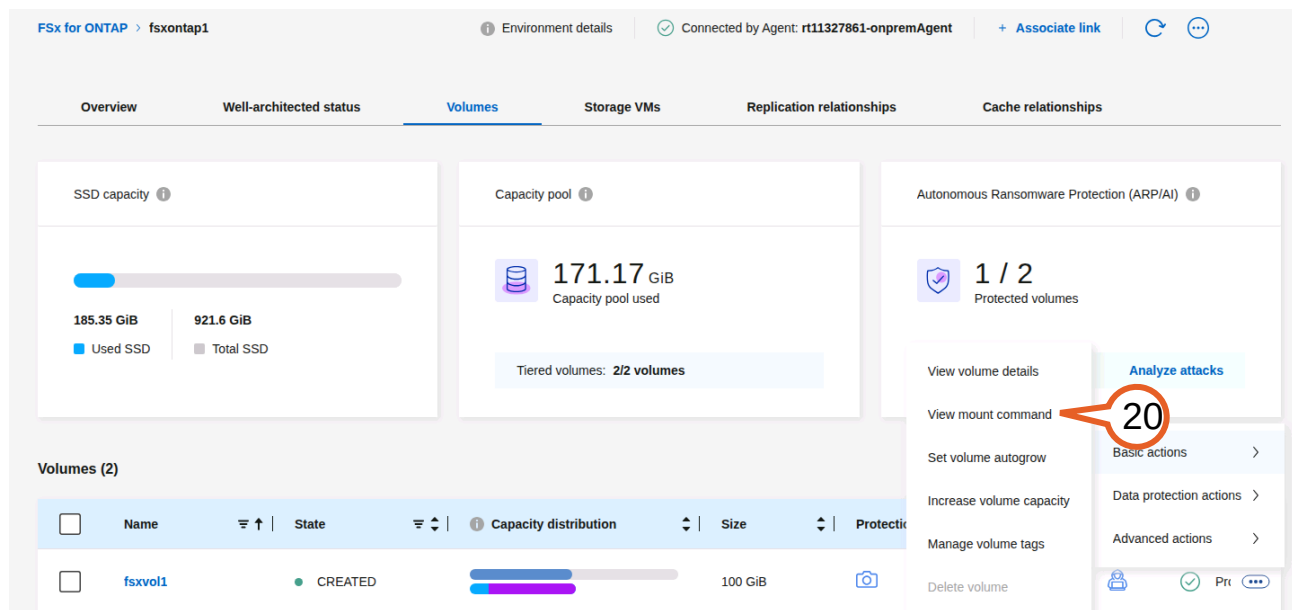


Figure 2-37:

21. The Linux command to mount this new NFS export displays, and a Copy button is presented to quickly copy this command to the clipboard to use in configuring the Linux client. You will not be accessing a Linux client in this demonstration, so click **Close**.

Mount command

To mount the volume to the mount point, copy the following command and enter it in the Linux terminal.

```
mount svm-99da0d62a3e40c5b0.fs-ac0fe80c69a907.fsx.us-east-1.amazo...
```



Close

Figure 2-38:

In this activity, using a single wizard in the NetApp Workloads console, you created a new FSx for ONTAP volume using the NFS protocol with both ONTAP storage efficiency and storage tiering features enabled. Next you will explore some of the most valuable data management features for data protection including snapshotting, cloning, and backups.

2.3.2 Explore Data Management Features

NetApp Console lets you use powerful ONTAP features like Snapshot and FlexClone technologies with your Amazon FSx for NetApp ONTAP storage.

- **Snapshot copies** provide a high-performance and space-efficient point-in-time copy of your volume. Snapshot copies are useful to protect against accidental or malicious data loss in your environment. When taking a snapshot of a volume, you are not duplicating the data, but preserving the data as it is at the time of the snapshot. With this efficient use of space and no performance impact, ONTAP systems are able to retain 1023 Snapshots of a volume. This, combined with the fact that taking a Snapshot copy is nearly instantaneous, gives you tremendous flexibility in how you approach data protection from ransomware, or any other form of data loss.
- **FlexClone technology** allows you to make fast and lightweight writable clones of your volumes. Similar to Snapshot technology, FlexClone volumes are point-in-time copies of a parent volume and are space-efficient because they share the same data blocks with their parent volume for common data. With this, DevOps and SRE teams can accelerate CI/CD cycles by rapidly creating multiple copies of live production data to be used for development, testing, and staging environments.

There are a few backup options available when using AWS FSx for NetApp ONTAP. Each have their tradeoffs and can fit different needs.

- **AWS Backup:** From AWS Console, AWS Backup is a backup service provided by AWS, enabling general backups for multiple services across the AWS offerings, but it does not take full advantage of specific features within ONTAP. While NetApp Workload Factory is not able to manage AWS backup, it provides an indicator whether the service is enabled in the Protection column and in the Data Protection view of the Storage dashboard. <https://aws.amazon.com/backup/>
- **FSx Backup for NetApp ONTAP:** Another AWS Console backup option is FSx Backup. This can be configured and managed from NetApp Workload Factory and is known as FSx for ONTAP backup. FSx Backup is set at the file system level and offers volume level protection, but it is also ONTAP aware for FSx for NetApp ONTAP file systems. Manual backups and restores can be performed from NetApp Workload Factory for each of the volumes in the file system using Snapshot technology in ONTAP. ONTAP is used to move snapshots to object storage, volume level backups taken once a day or manual and they give more control for labeling, restoring, ARP/AI, backup targets to different regions, immutability, scheduling. <https://docs.aws.amazon.com/fsx/latest/ONTAPGuide/using-backups.html>
- **NetApp Backup and Recovery:** From NetApp Console, NetApp Backup and Recovery is also an option as NetApp Backup and Recovery does support MSSQL on FSxN. NetApp Workload Factory is also aware of the data service and does offer redirection for Microsoft Databases for faster onboarding. Workload Factory automatically registers your MS SQL server host for you and installs the necessary plugins to get started with application consistent data protection. The Backup and Recovery Data Service, when used with MS SQL servers with FSxN, give most of the same ONTAP functionality with NetApp Workloads FSx Backup, but adds application consistent backups with an enterprise-wide management hosted service. <https://docs.netapp.com/us-en/bluexp-backup-recovery/br-use-mssql-protect-overview.html>



Note: Backup and Recovery **does not support** Volume level backup with FSx for NetApp ONTAP. <https://docs.netapp.com/us-en/bluexp-backup-recovery/prev-ontap-protect-overview.html#supported-bac...>

- **NetApp SnapCenter:** SnapCenter is a standalone server that supports application consistent backups running on ONTAP storage. Amazon FSx for NetApp ONTAP is a supported storage system with SnapCenter. https://docs.netapp.com/us-en/snapcenter/get-started/reference_supported_storage_systems_and_applications.html#supported-storage-systems Users who start in SnapCenter can now migrate to the hosted data service NetApp Backup and Recovery very easily by importing their SnapCenter configurations directly in Backup and Recovery. <https://docs.netapp.com/us-en/bluexp-backup-recovery/br-start-discover.html#if-snapcenter-is-already-installed-import-snapcenter-resources-into-bluexp-backup-and-recovery>

In this activity, you will use NetApp Workloads to create a manual backup of the FSx for ONTAP backup service from AWS. Then you create a volume-level snapshot copy and use FlexClone technology to create a clone of an FSx for ONTAP volume.

1. You should already be at the Volumes tab within the fsxontap working environment and scrolled down to the list of volumes at the bottom. Click the **three horizontal dots** > **Data protection actions** > **FSx for ONTAP Backup** > **Manual backup** from the menu.

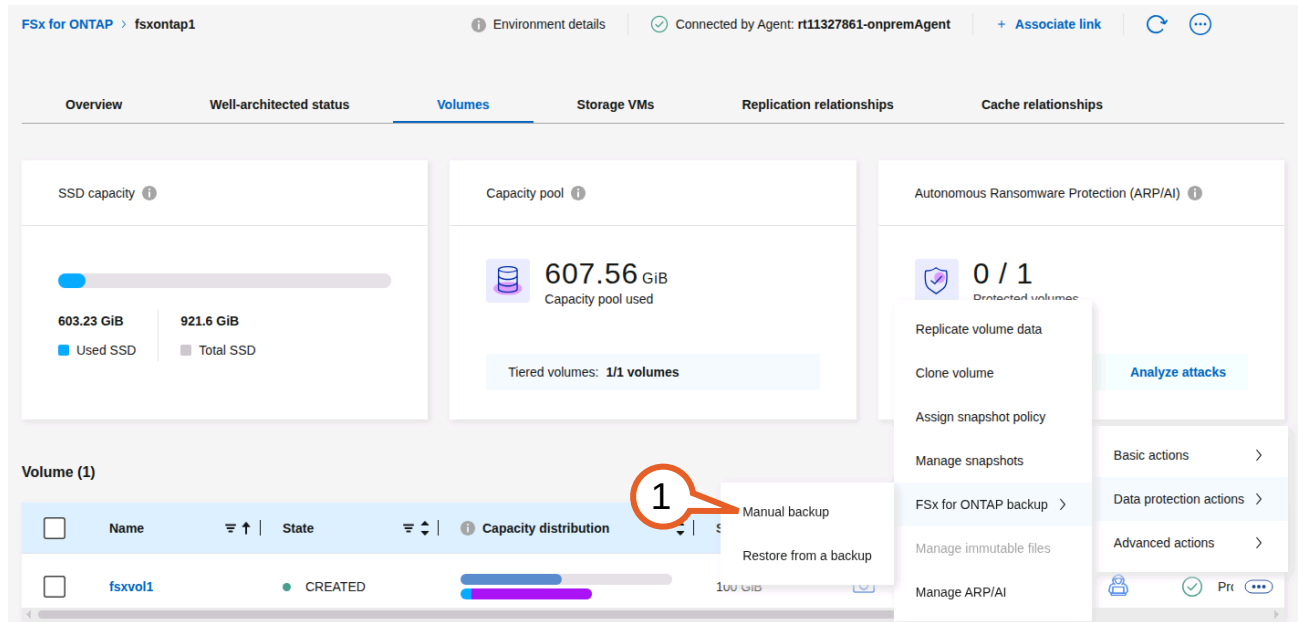


Figure 2-39:

2. Enter a name for your manual backup, or keep the default and click **Backup**.

Manual backup

This manual backup is independent from your scheduled backups.

Backup name

fsxvol1_backup_2025-10-01T22:03:21.262Z



Figure 2-40:

3. Creating Manual backup outside of your scheduled ones is very quick.

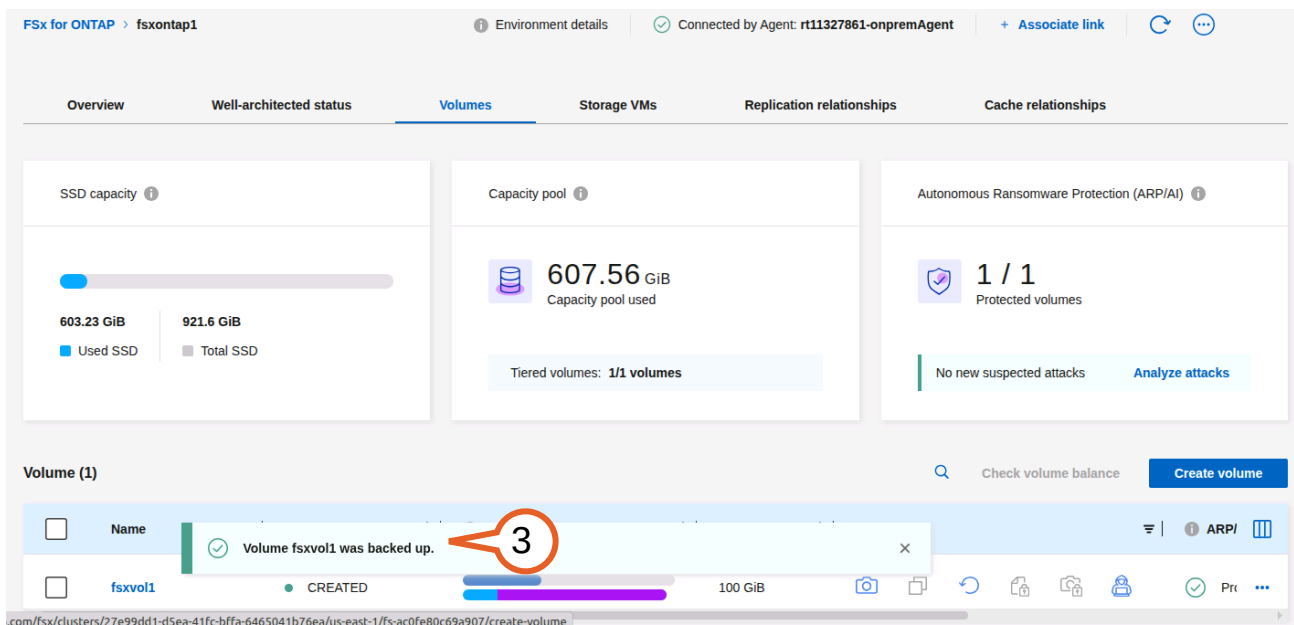


Figure 2-41:

4. You should already be at the Volumes tab within the fsxontap working environment and scrolled down to the list of volumes at the bottom. Click the **three horizontal dots** > **Data protection actions** > **Manage snapshots** from the menu.



Note: Notice, but do not select, the “Restore volume from a snapshot” option that allows you to recover lost data from prior snapshot copies taken either from the snapshot policy schedule, or manual snapshot copies like the one you are creating now.

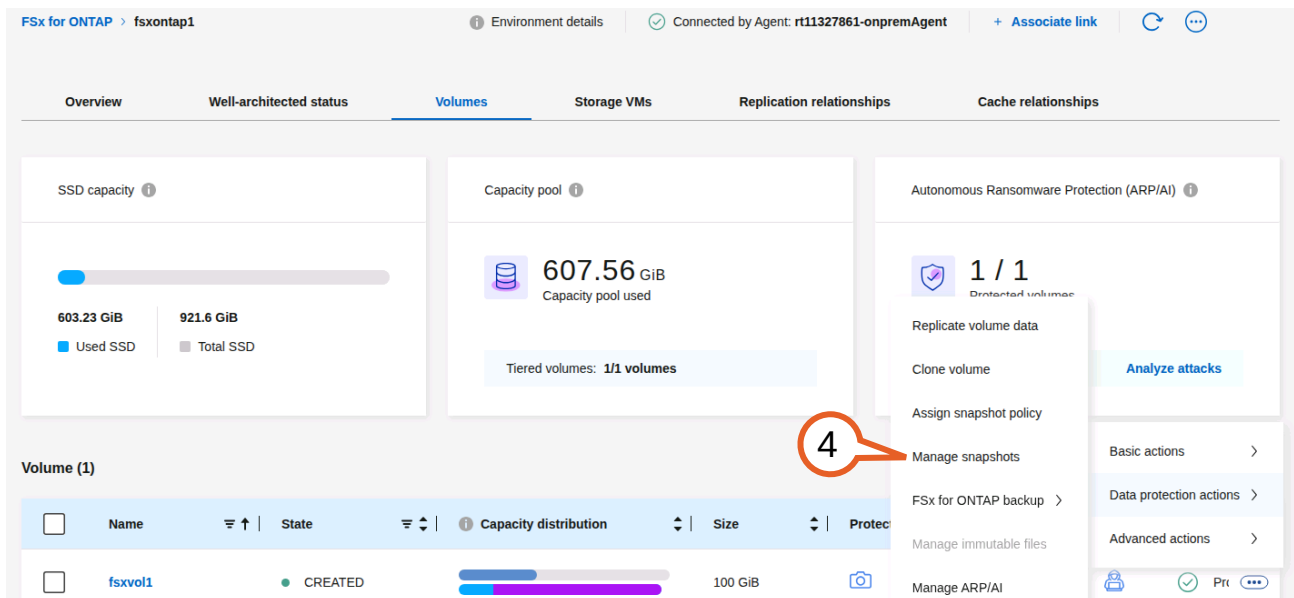


Figure 2-42:

5. Click **Create snapshot**.

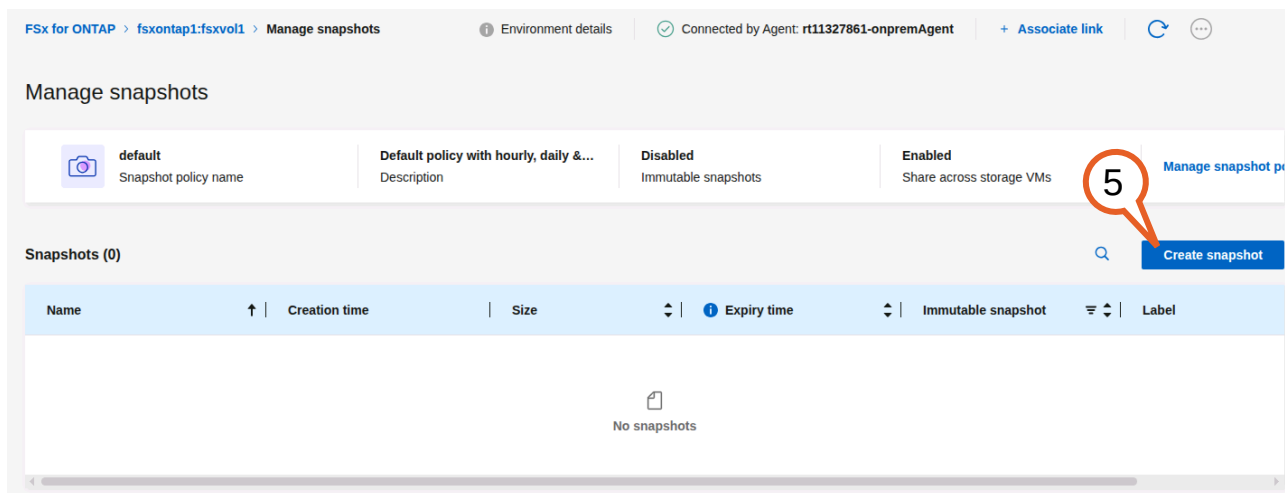


Figure 2-43:

6. A snapshot is a read-only image of the FSx for ONTAP volume at a time. Using this operation you can give it a custom name, add a label, set a retention period and if desired, set the snapshot to be immutable which prevents it from accidental deletion.
7. Click **Create**.



Tip: You can store up to 1023 snapshot copies per file system at any point in time. When you reach this limit the next snapshot copy replaces the oldest snapshot copy. You can also set a retention period or even

make the snapshot immutable from this one action. Immutable snapshots unchangeable copies of your data at a specific point in time meant to provide added protection.

Create a snapshot

A snapshot is a read-only image of an FSx for ONTAP volume at a time.

Snapshot name

Label

manually.2025-10-01_2204

daily

Retention period

30

days

Expiry time: **October 31, 2025 at 10:05 PM**

☒ **Make this snapshot immutable**

Protect snapshot data against malicious acts and accidental deletion. Powered by ONTAP Tamperproof Snapshots.

☒ I understand that immutable snapshots and the file system they are hosted on are protected during the retention period and can't be deleted until the retention period ends.

Create

Cancel

Figure 2-44:

8. The snapshot now appears in the Manage snapshots view. Return back to the volume view by clicking the **volume name** at the top.

FSx for ONTAP > fsxontap1:fsxvol1 > Manage snapshots

Environment details

Connected by Agent: rt11327861-onpremAgent

Associate link

Manage snapshots

default	Default policy with hourly, daily &...	Disabled	Enabled	Manage snapshot p...
Snapshot policy name	Description	Immutable snapshots	Share across storage VMs	

Snapshot (1)

Name	Creation time	Size	Expiry time	Immutable snapshot	Label
manually.2025-10-01_2204	Oct 1, 2025, 10:06 PM	37.17 KiB	Oct 31, 2025, 10:06 PM	Protected	daily

Figure 2-45:

9. Click the **three horizontal dots** > **Data protection actions** > **Clone volume** from the menu.

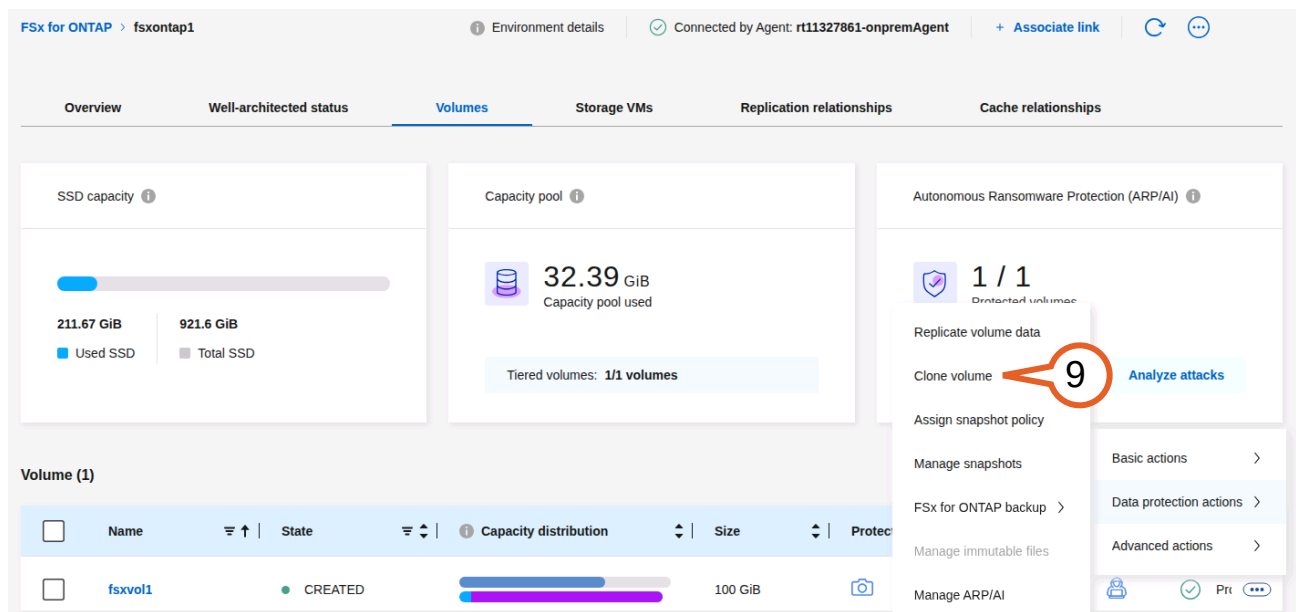


Figure 2-46:

10. Click **Clone**.

Clone volume

Cloning the volume creates a read-write volume from the original volume.

Clone volume name

fsxvol1_clone



Figure 2-47:

11. The newly created clone volume now appears in the volume list. The cloned volume shares the same capacity, usage, and tiering policy as the parent volume.

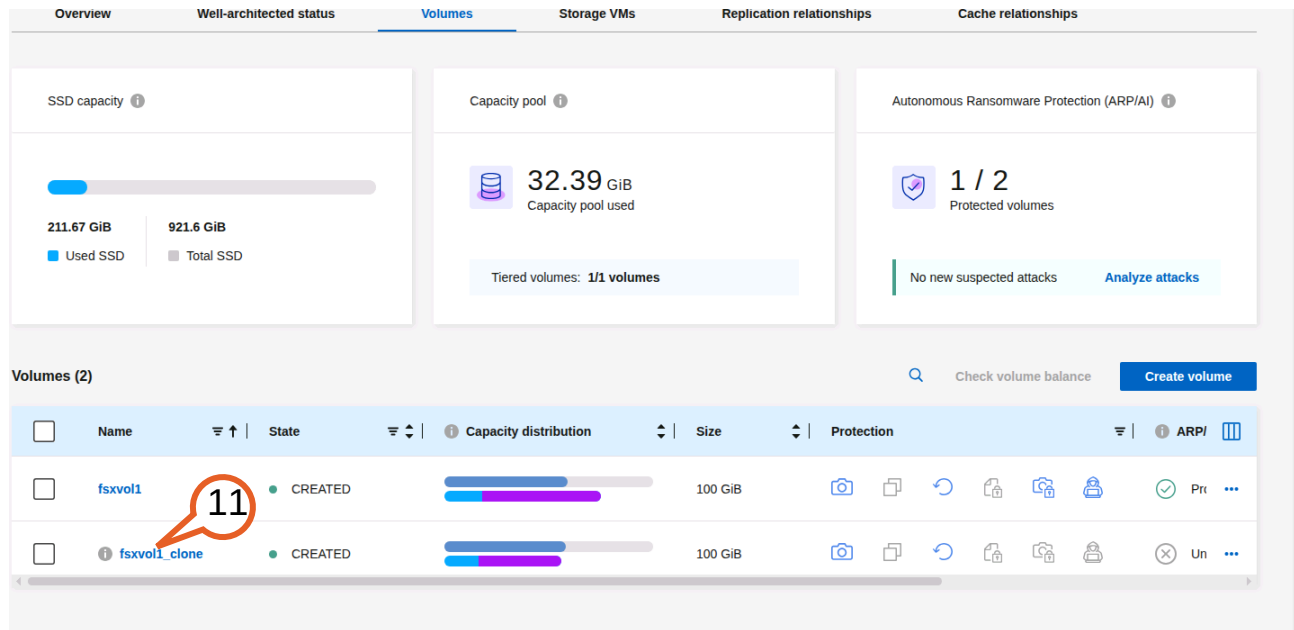


Figure 2-48:

Takeaway: NetApp Workloads allows storage administrators to quickly and easily take advantage of Amazon FSx for NetApp ONTAP. Despite offering a wide breadth of features and protocols unique to a fully managed cloud storage service, provisioning and management is simple and can be done in minutes. Using NetApp Workloads offers features not directly available in the AWS Console for AWS FSx for NetApp ONTAP customers.

2.4 Data Replication with Amazon FSx for NetApp ONTAP

Data replication is a significant concern for companies with valuable data, and managing backups and long-term data retention on premises can be complex, time consuming, and costly. To serve this need Amazon FSx for NetApp ONTAP allows storage administrators to easily back up on-premises data hosted on ONTAP clusters to the cloud (or vice-versa) without the need to re-engineer applications or disrupt operation.

FSx for ONTAP offers centralized, policy-based Snapshot, SnapMirror (which enables disaster recovery with a single recovery point), and SnapVault (a backup with multiple recovery points) to easily backup, archive, or replicate on-premises workloads into the Cloud. FlexCache volumes are also available. With the NetApp Console, administrators can configure SnapMirror or SnapVault data protection relationships between on-premises volumes and FSx for ONTAP file systems.

In this activity, you setup a data protection relationship between two FSx for ONTAP file systems.



Important: This part of the demonstration requires steps from the previous section to be completed successfully. At a minimum, an FSx for ONTAP instance has to be created and at least one FSx for ONTAP volume must be created.

1. Click **Storage > Management**.

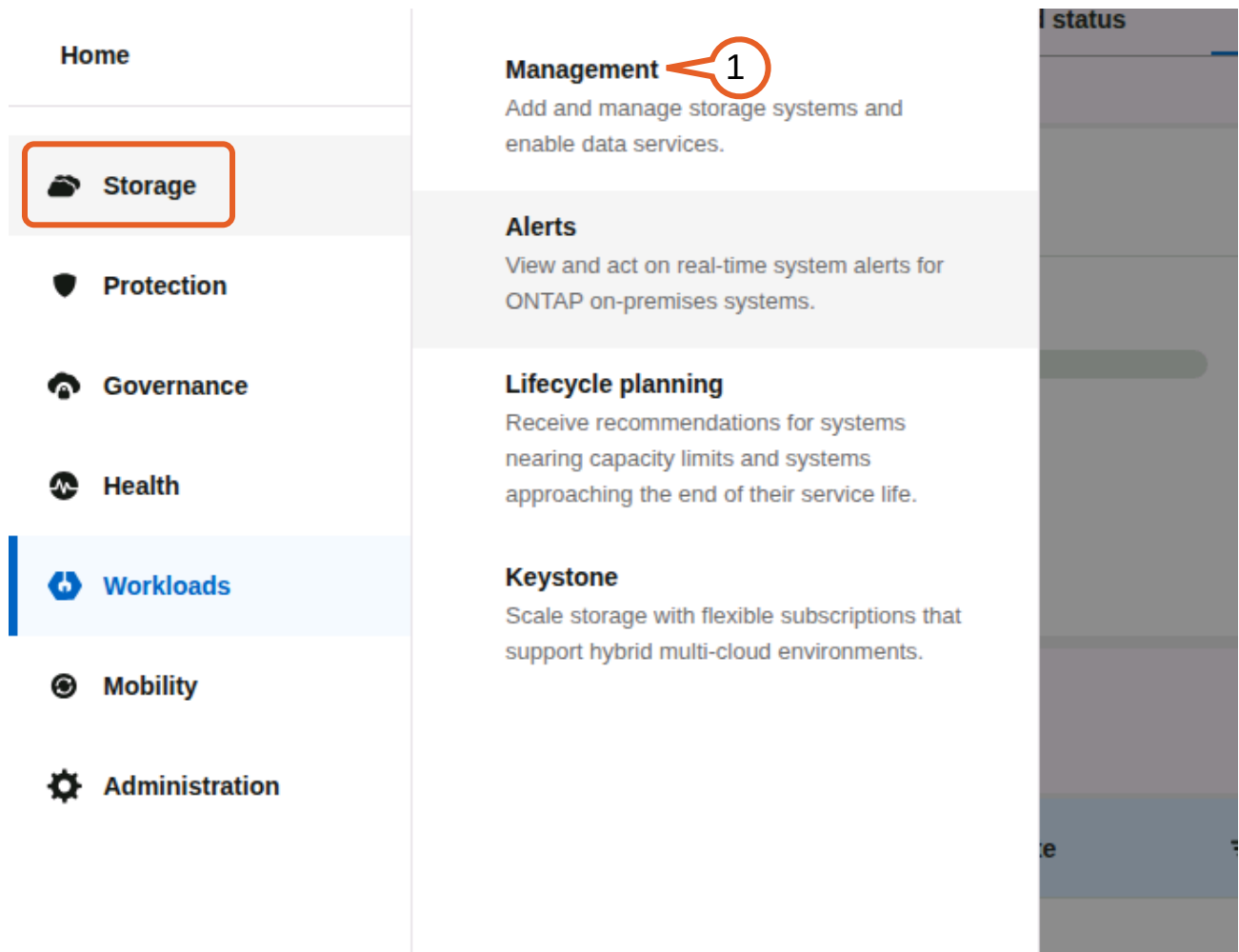


Figure 2-49:

2. Notice the volume count on both file systems. You will be migrating data from a volume in fsxontap1 to fsxontap2.

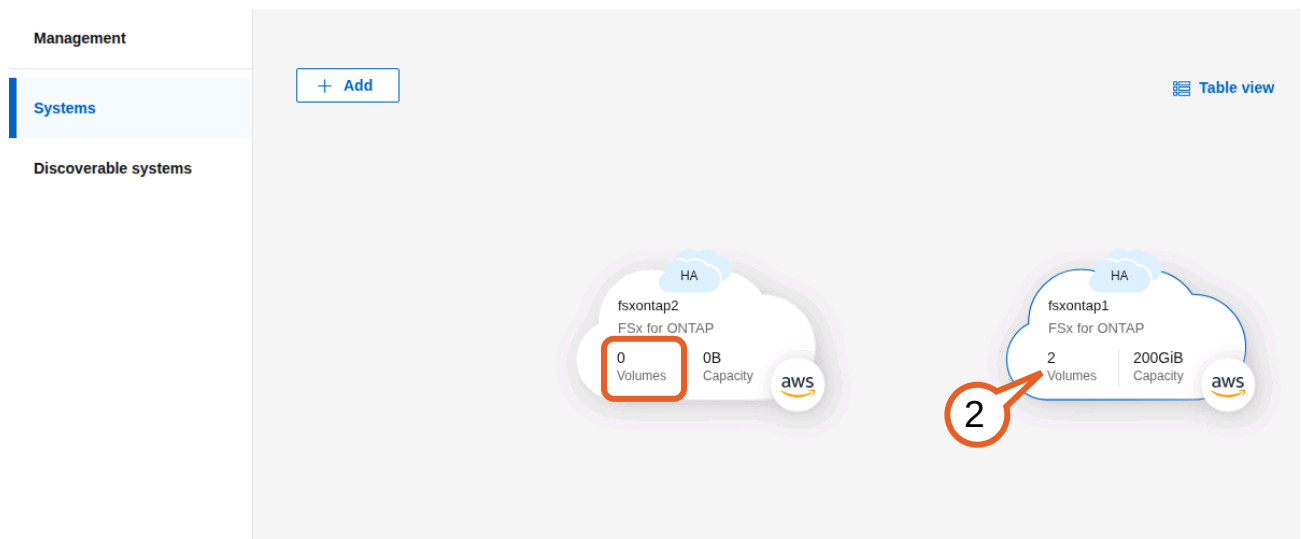


Figure 2-50:

- Return to the Systems Management view to see the file system you just created, and drag **fsxontap1** to the **fsxontap2** file system.



Note: Remember replication can happen in either direction between an on-premises system and an FSx for NetApp ONTAP file system, or between two cloud instances of FSx for NetApp ONTAP even in different AWS Availability Zones around the globe. With proper networking you can even set up replication with CVO instances running anywhere.

- Select **Replication**.



Note: Volume Caching will create a NetApp FlexCache volume that enables read-intensive workflows to copy data from on-prem to FSx for ONTAP, vice versa, or between two FSx for ONTAP file systems. These copies are fast and space-efficient reducing WAN costs by only consuming a fraction of the story capacity and no sending over a full data copy. Deployments with multiple AWS FSx for ONTAP instances with Volume Caching enabled can move large datasets for global collaboration with consistency in fast and bandwidth efficient copies with lower latency. Feel free to explore enabling caching in addition to replication.

This will create a replication relationship between the two environments. Because you dragged from fsxontap2 to fsxontap1, the first file system will act as the source ONTAP cluster while the fsxontap2 file system is the destination.

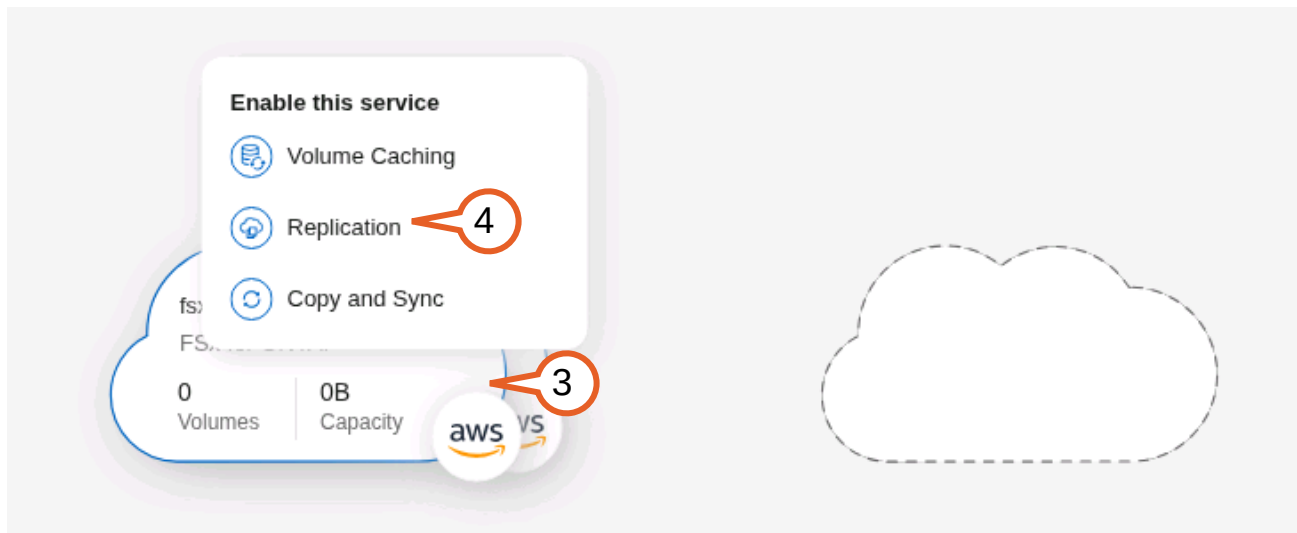


Figure 2-51:

- Select **fsxvol1**.

6. Click **Next**.

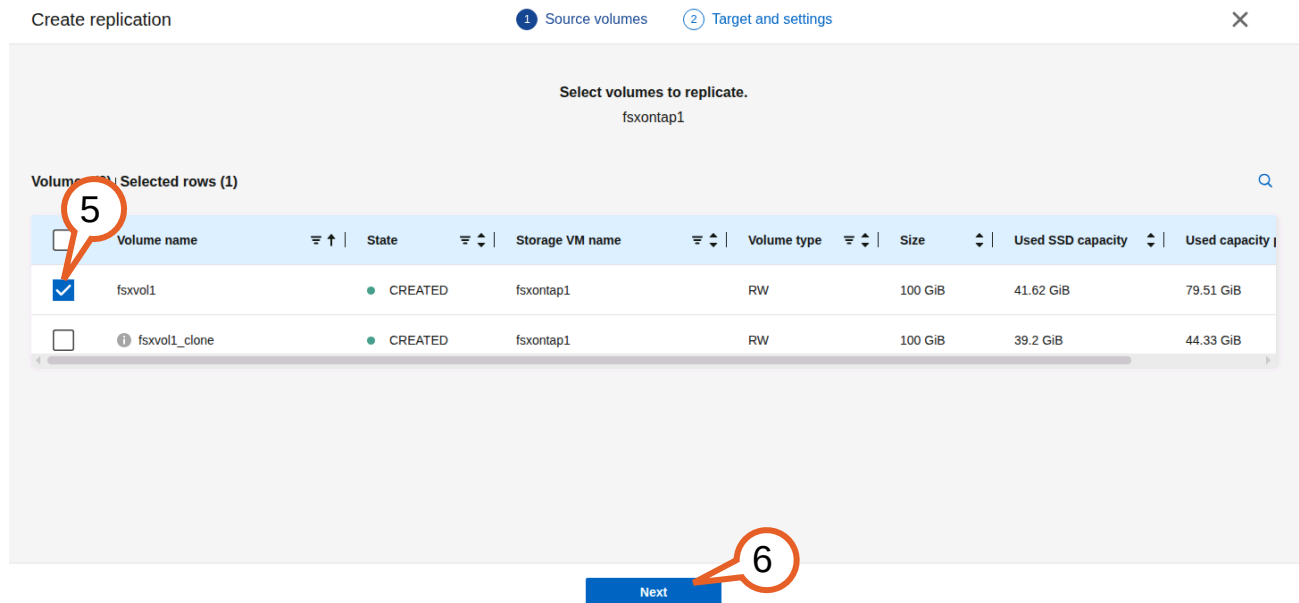


Figure 2-52:

7. Scroll down the options in Create replication.

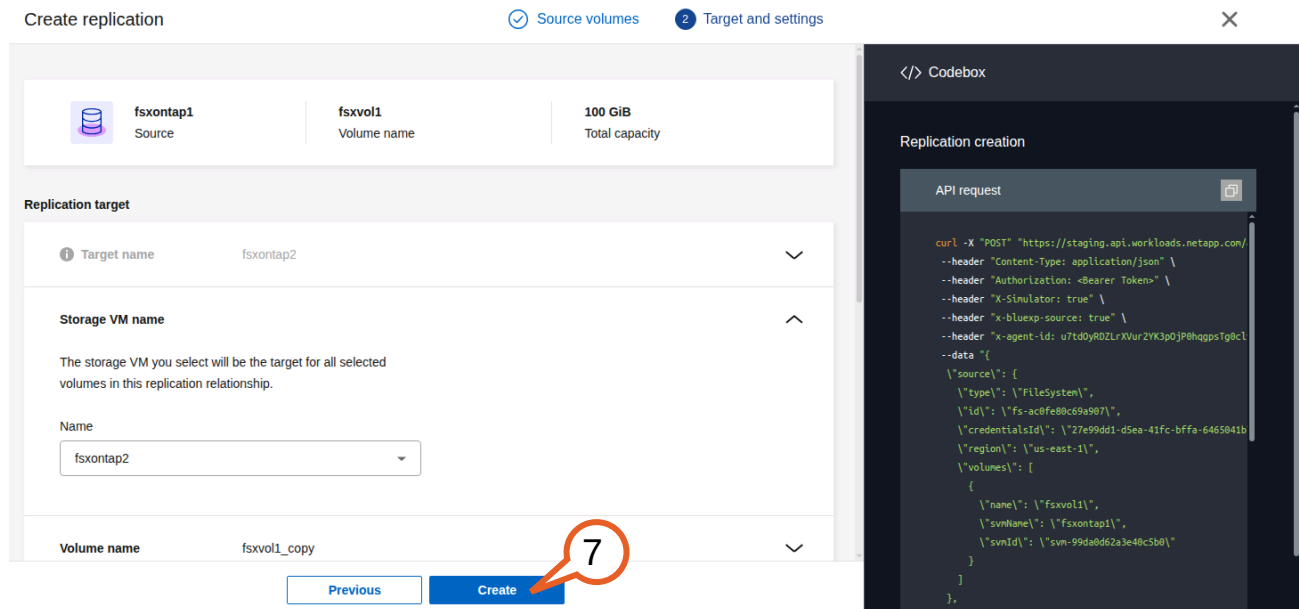


Figure 2-53:

8. Expand the Use case section to see the options for replication and click **Migration**.

9. Click **Create**.

Create replication ✓ Source volumes 2 Target and settings ✕

Use case

Select the use case for your replication.

☒ **Migration** 8

Seamlessly transfer your data to the selected FSx for ONTAP file system.

☐ **Hot disaster recovery**

Maintain real-time replication of critical workloads and ensure high availability and rapid disaster recovery.

☐ **Cold disaster recovery**

Cold: Cost-effective replication for disaster recovery with longer recovery time objectives (RTOs) and recovery point objectives (RPOs).
Archive: Replicate data for long-term storage and compliance.

☐ **Other**

Select the required configuration for your use case.

Previous Create 9

Codebox

Replication creation

API request

```
curl -X "POST" "https://staging.api.workloads.netapp.com/"
--header "Content-Type: application/json" \
--header "Authorization: <Bearer Token>" \
--header "X-Simulator: true" \
--header "x-bluexp-source: true" \
--header "x-agent-id: u7td0yR0ZLrXVurZYK3p0JP0hqqps1g0c1"
--data "{
  \"source\": {
    \"type\": \"FileSystem\",
    \"id\": \"fs-ac0fe80c63a9071\",
    \"credentialsId\": \"27e99dd1-d5ea-41fc-bffa-6465841b\",
    \"region\": \"us-east-1\",
    \"volumes\": [
      {
        \"name\": \"fsxvol11\",
        \"svnName\": \"fsxontap1\",
        \"svnId\": \"svn-99da062a3e40c5b0\"
      }
    ]
  },
  \"target\": {
    \"type\": \"FileSystem\",
    \"id\": \"fs-ac0fe80c63a9071\",
    \"credentialsId\": \"27e99dd1-d5ea-41fc-bffa-6465841b\",
    \"region\": \"us-east-1\",
    \"volumes\": [
      {
        \"name\": \"fsxvol11\",
        \"svnName\": \"fsxontap1\",
        \"svnId\": \"svn-99da062a3e40c5b0\"
      }
    ]
  }
}
```

Figure 2-54:

10. Because Replication creation is in progress, it is not visible yet.

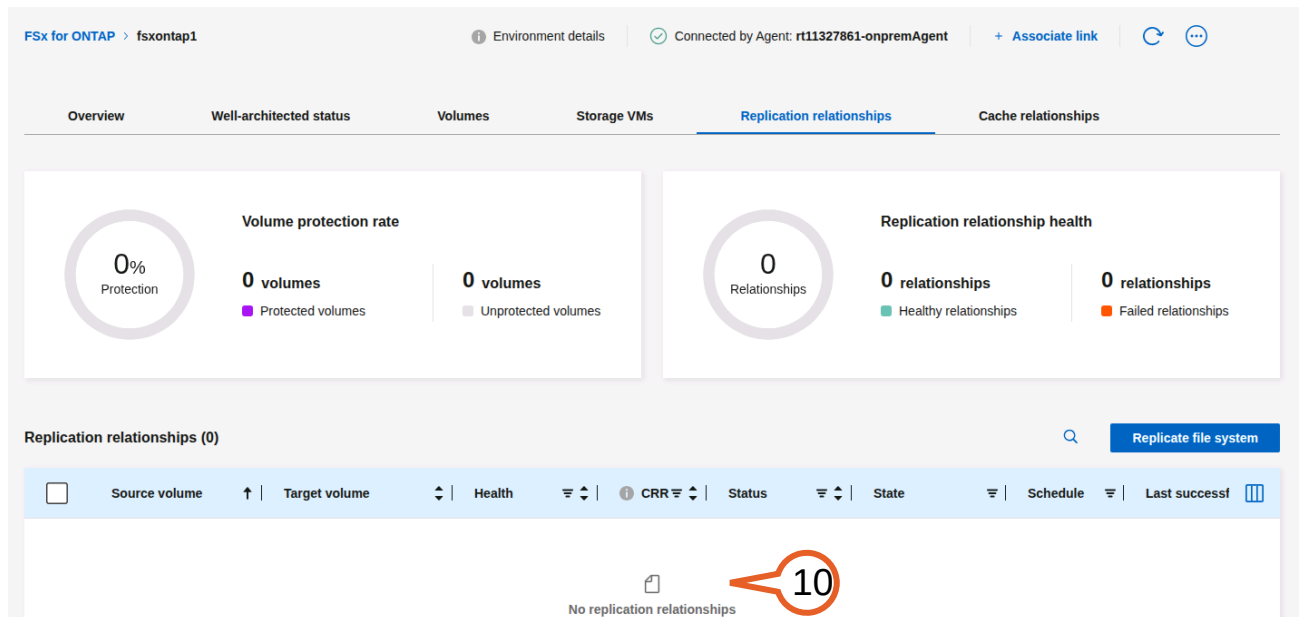


Figure 2-55:

11. Return the top level three horizontal line Home Menu, then to **Storage > Management > Systems** and wait for the Volumes count to change to 1.

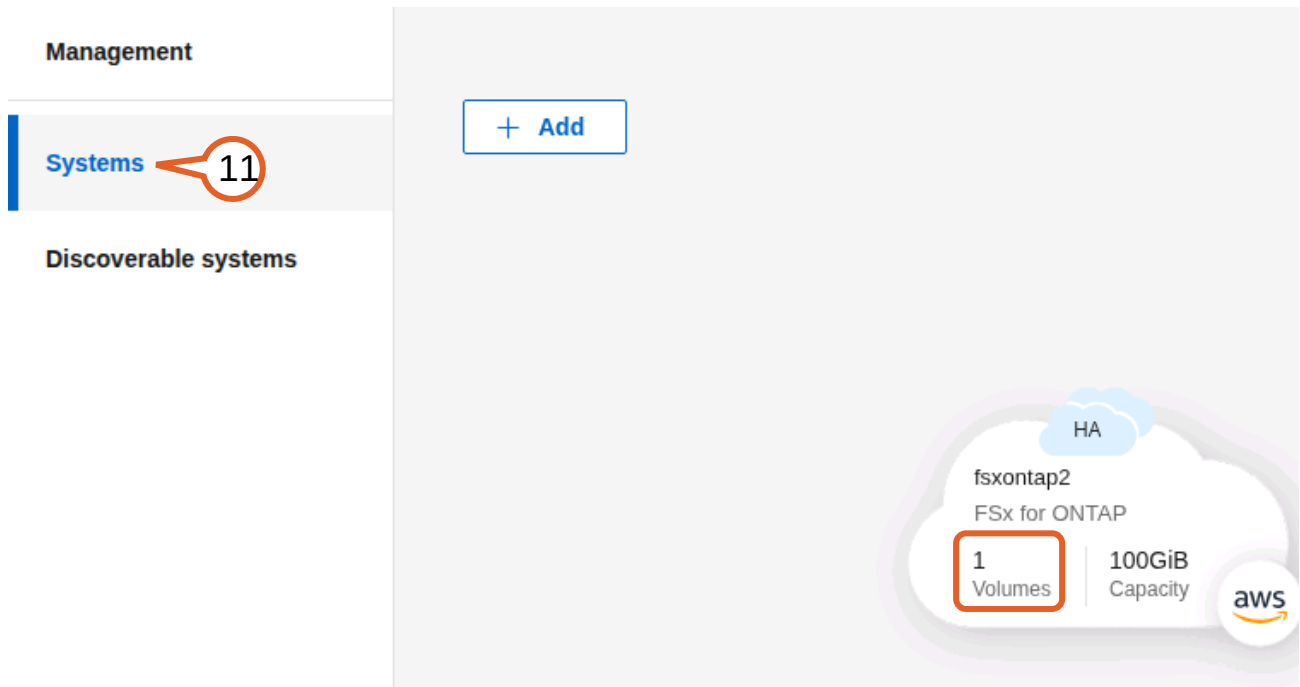


Figure 2-56:

12. After the volume appears on fsxontap2 return to **Workloads > Storage > FSx for ONTAP**.
13. Click the name **fsxontap2**.

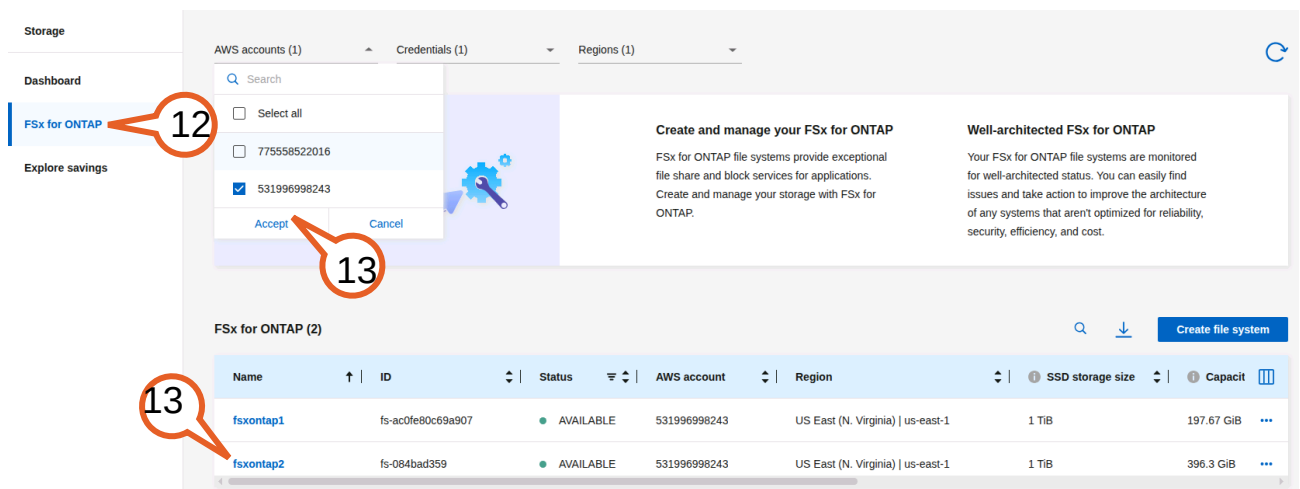


Figure 2-57:

14. The Overview will display “1” Replication relationship.

15. Click the **Replication relationships** tab.

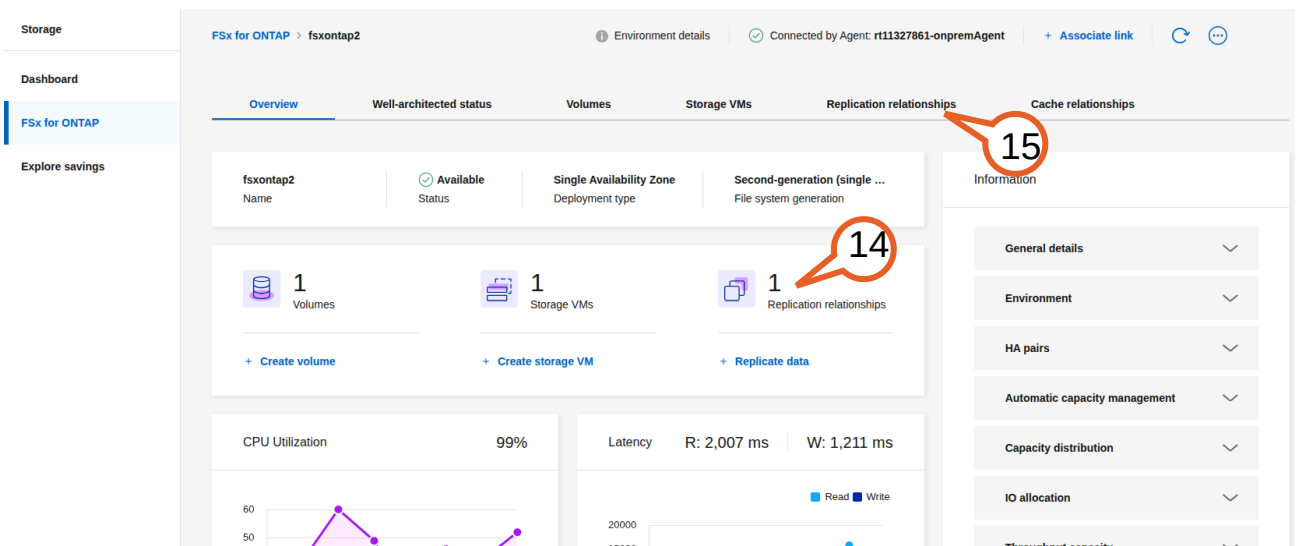


Figure 2-58:

16. The Replication relationship should have a status of “Healthy”.

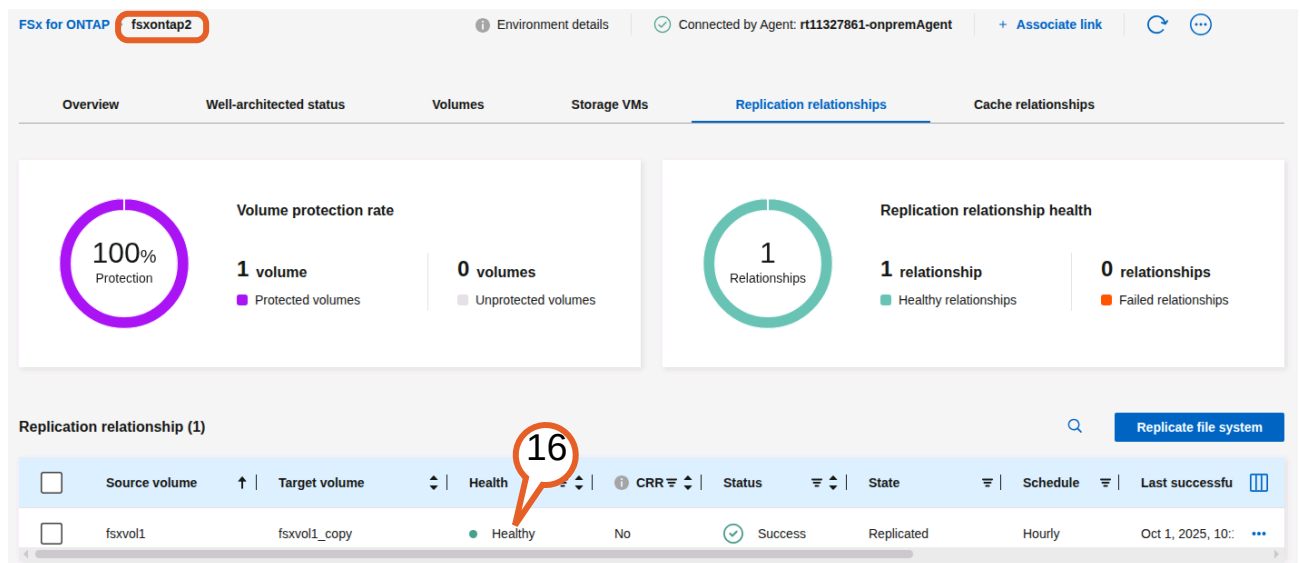


Figure 2-59:

17. Select the ... button to the right of the replication relationship.

18. Click **Break**.



Note: The replication relationship transfers the data to a read only volume at the destination, but it can be updated until the data migration is ready to be completed.

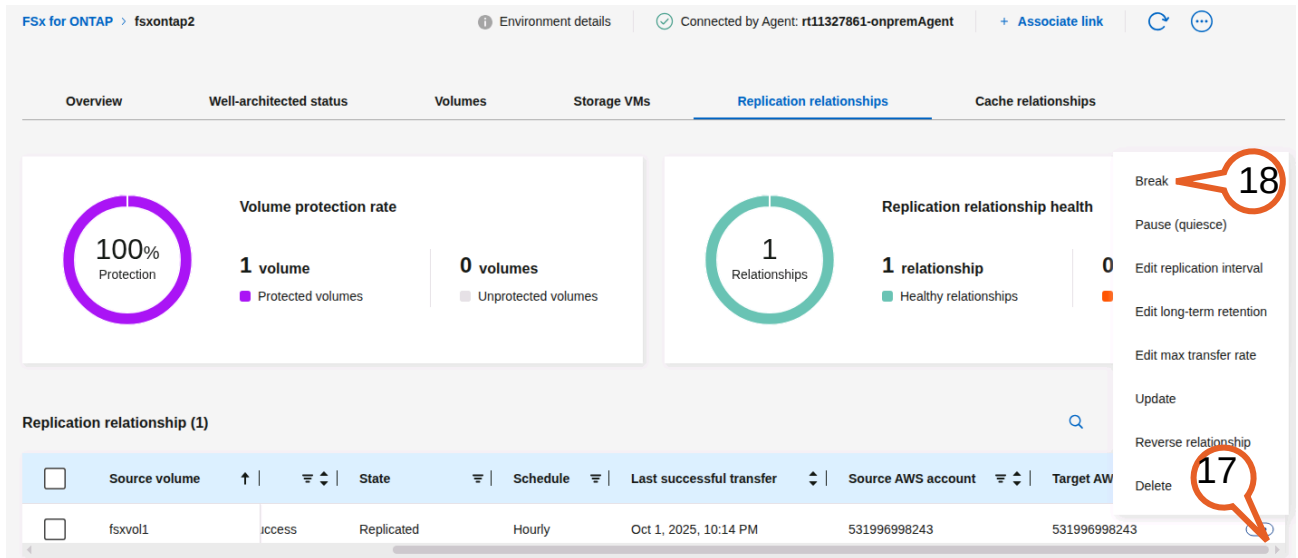


Figure 2-60:

19. Click **Break**.

Break relationship

Are you sure that you want to break the relationship between **fsxvol1** and **fsxvol1_copy**?



Figure 2-61:

20. Go back to the options menu, click **Delete**.

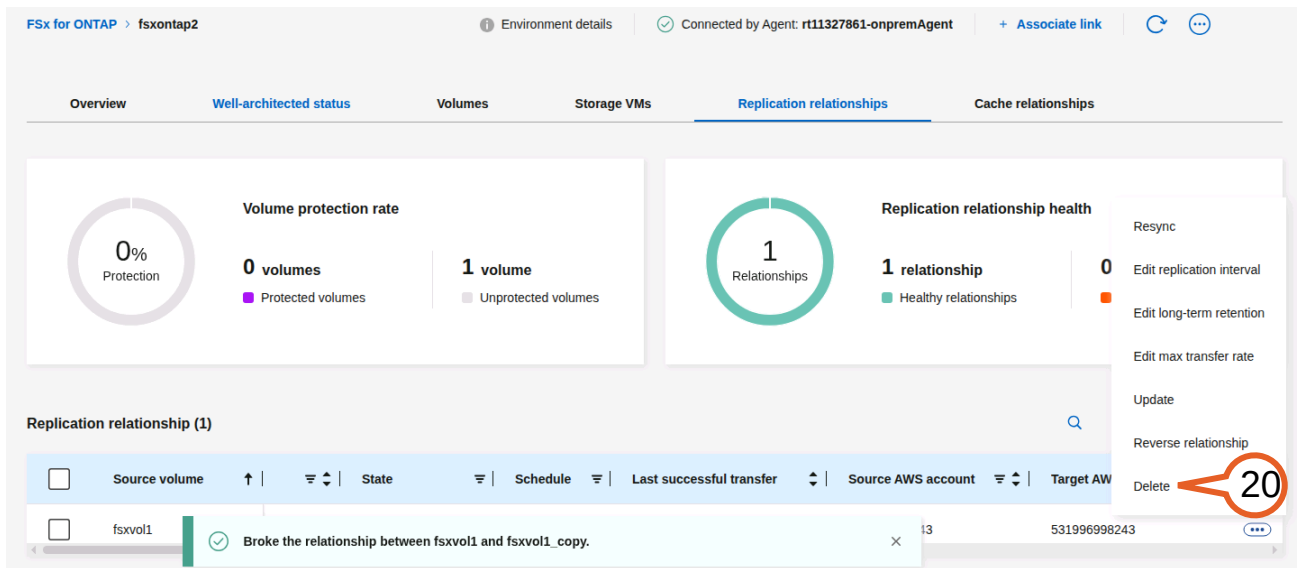


Figure 2-62:

21. Click **Delete**.

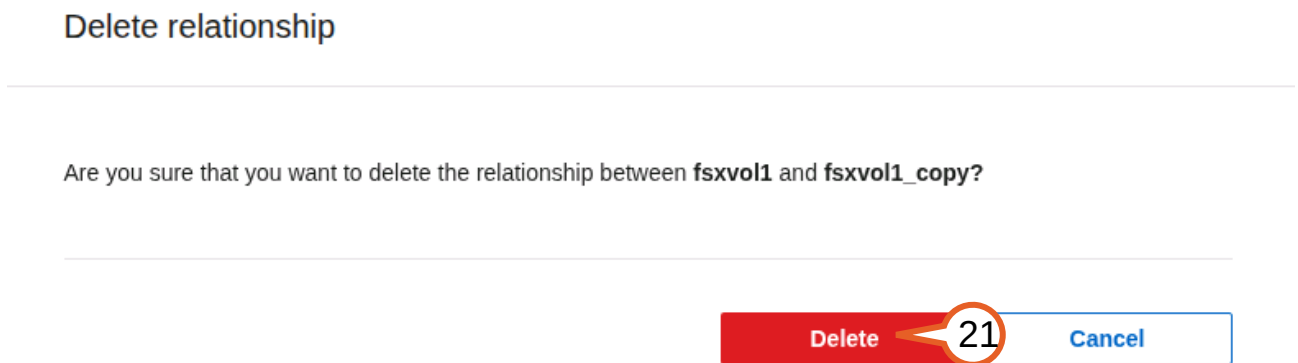


Figure 2-63:

22. Click on the notification at the top to expand out the message.

23. Click **Audit**.

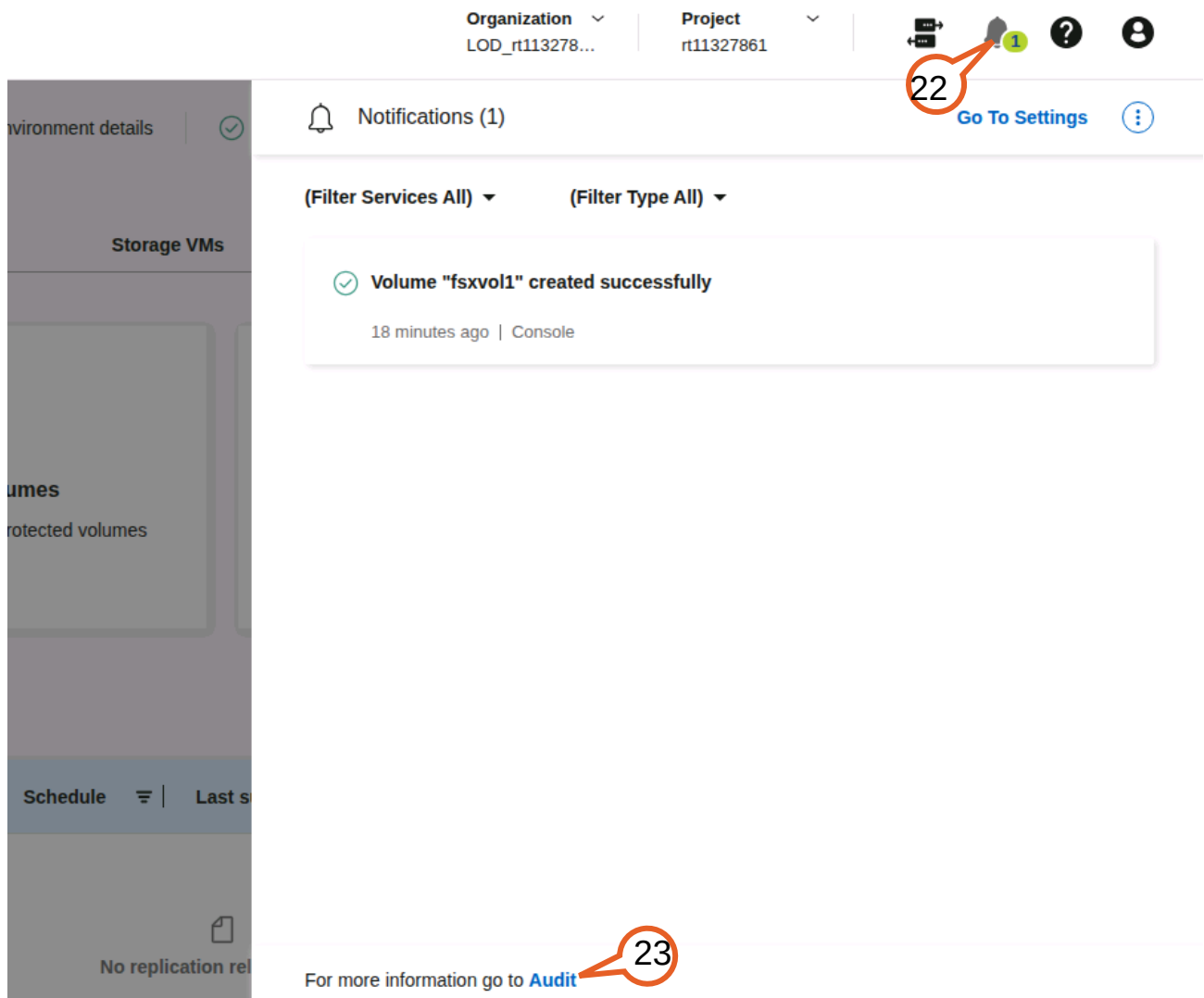
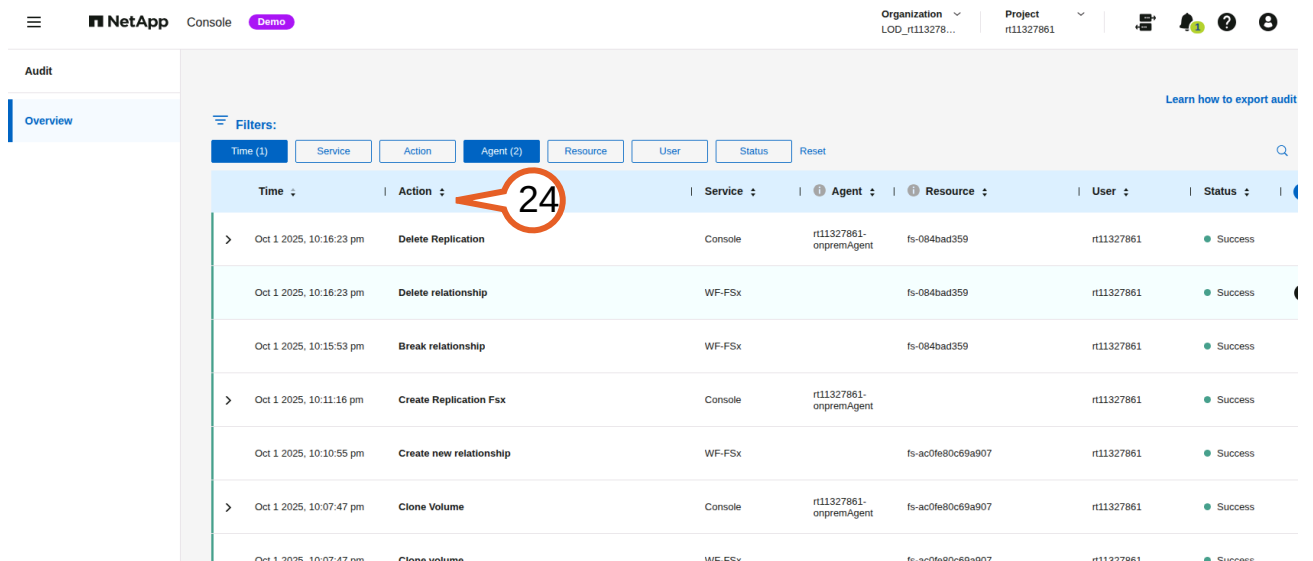


Figure 2-64:

24. The Audit Overview gives a detailed timeline of all actions completed.



NetApp Console Demo

Organization: LOD_rt11327861... Project: rt11327861

Audit Overview

Filters: Time (1) Service Action Agent (2) Resource User Status Reset

Learn how to export audit

Time	Action	Service	Agent	Resource	User	Status
> Oct 1 2025, 10:16:23 pm	Delete Replication	Console	rt11327861-onpremAgent	fs-084bad359	rt11327861	Success
Oct 1 2025, 10:16:23 pm	Delete relationship	WF-FSx		fs-084bad359	rt11327861	Success
Oct 1 2025, 10:15:53 pm	Break relationship	WF-FSx		fs-084bad359	rt11327861	Success
> Oct 1 2025, 10:11:16 pm	Create Replication Fsx	Console	rt11327861-onpremAgent		rt11327861	Success
Oct 1 2025, 10:10:55 pm	Create new relationship	WF-FSx		fs-ac0fe80c69a907	rt11327861	Success
> Oct 1 2025, 10:07:47 pm	Clone Volume	Console	rt11327861-onpremAgent	fs-ac0fe80c69a907	rt11327861	Success
Oct 1 2025, 10:07:47 am	Clone volume	WF-FSx		fs-ac0fe80c69a907	rt11327861	Success

Figure 2-65:

25. Return to **Workloads > Storage**.

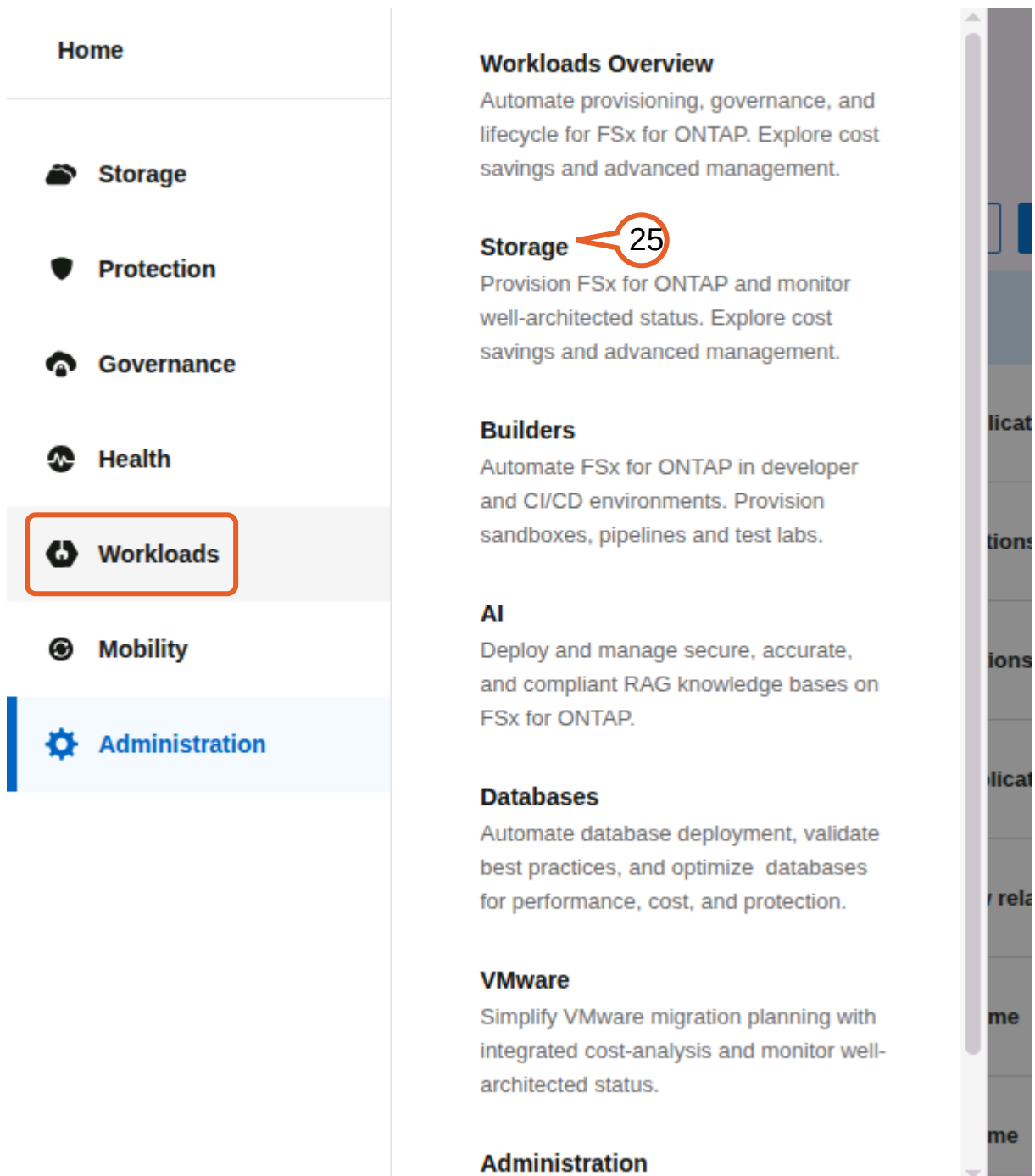


Figure 2-66:

26. Make your way to the Volumes tab for fsxontap2.

27. Click the **more (three buttons) icon** > **Basic actions** > **View volume details**.

The screenshot shows the NetApp console interface for 'FSx for ONTAP'. The breadcrumb navigation at the top is 'FSx for ONTAP > fsxontap2', with a callout '26' pointing to 'fsxontap2'. The 'Volumes' tab is selected. On the right, the 'Basic actions' menu is open, and a callout '27' points to the 'View volume details' option. The main content area shows SSD capacity (632.07 GiB used, 921.6 GiB total), a capacity pool of 72.14 GiB, and a table of volumes. The table has one entry: 'fsxvol1_copy' with a state of 'CREATED' and a size of 100 GiB.

Figure 2-67:

28. Notice the Volume type is now “Read-Write (RW)”. You now have a volume migration that has completed and the new destination volume sitting in the destination file systems is an independent copy of the source data.

The screenshot shows the 'View volume details' page for the volume 'fsxvol1_copy'. The breadcrumb navigation is 'FSx for ONTAP > fsxontap2:fsxvol1_copy > View volume details'. The page displays various details about the volume, including its name, UUID, ID, creation time, state, and type. A callout '28' points to the 'RW' (Read-Write) volume type. The 'Tags' section shows '1 - Manage volume tags'. A 'Close' button is at the bottom.

General details	Distribution	Consumption	Access	Protection
Volume name	fsxvol1_copy			
Volume UUID	14e8569c-b1ca-4491-ae08-036c69fd82a9			
Volume ID	fsvol-196a58c36a312d9ab			
Creation time	Wed, Oct 1, 2025, 10:10 PM			
State	Created			
Volume type	RW			
Hosting SVM	fsxontap2			
Tags	1 - Manage volume tags			

Figure 2-68:

Using ONTAP volume snapshots and their easily created replication or caching relationships, storage administrators can protect against loss of data, ransomware attacks, achieve long-term data retention, and even update large datasets globally with consistency and efficiency. NetApp Console enables organizations to easily back up or move data copies to the cloud using SnapMirror for FlexCache technology, meaning the replication is efficient and non-disruptive.

3 Lab Summary

Amazon FSx for NetApp ONTAP brings the full capabilities of ONTAP to a native AWS managed service to easily power primary workloads and quickly build modern applications. Through NetApp Console, NetApp Workloads combines these enterprise-grade capabilities with the simplicity and turn-key operation commonly found with managed storage services that are less feature-rich.

Amazon FSx for NetApp ONTAP is the only AWS managed storage option that can serve both SMB/CIFS, NFS, and NVMe storage protocols, and it is the only Amazon FSx service available with support for iSCSI. This versatility means you can access a volume from variety of clients (Linux, Windows, or macOS) using either a NAS or SAN-style configuration. You can even use NFS and SMB/CIFS to access the same volume concurrently. This extensive protocol support means the service works well with almost any workload.

Adding NetApp Workloads gives a hybrid cloud customer using FSx for ONTAP a free way to take advantage of NetApp value-added capabilities not available from the AWS Console. Getting started is easy and just requires a NetApp Console account and AWS Credentials. Workloads helps you find opportunities to better manage growing costs and implement best practices. Added features like Codebox lets you accelerate infrastructure-as-code for multiple formats: API requests, CloudFormation, and Terraform.

From the NetApp Workloads interface, a NetApp customer can take advantage of more Snapshot policy control, data tiering management, data tiering management, multi-protocol file sharing access, data protection visibility across workloads, Replication, simplified Block storage management, access to FlexClone technology, and automatic capacity management.

To further enhance NetApp options, a NetApp Console Agent can be easily deployed either on-premises, or in AWS, which enables access to try and subscribe for NetApp Data Services including the following:

- NetApp Backup and Recovery to backup applications like Microsoft SQL Server in an application consistent backup, and easily restore and clone database workloads.
- NetApp Data Classification to add additional governance and scan your data to ensure compliance.
- NetApp Disaster Recovery is especially helpful for VMware Cloud on AWS customers with FSx for ONTAP and VMware with ONTAP anywhere to protect VMs.
- NetApp Ransomware Resilience to provide additional visibility and awareness for ransomware protection for NAS and SAN workloads.

4 Additional Information

4.1 References

The following references were used to create this lab guide:

- [NetApp Console - Learn about Amazon FSx for NetApp ONTAP](#)
- [Amazon FSx for NetApp ONTAP management](#)
- [FSx for ONTAP User Guide](#)
 - [Managing FSx for ONTAP volumes \(contains information on tiering\)](#)
- [NetApp Workloads](#)

4.2 Version History

Version	Date	Document Version History
2.0.0	September 2024	updated version of Bluexp
2.2.0	February 2025	WLF updates
2.2.1	February 2025	Review feedback applied
2.3.0	March 2025	Updated BlueXP IAM in Std Mode
2.3.1	April 2025	Updated screenshot in BlueXP login
2.3.2	July 2025	Removed on-prem ONTAP discovery and SnapMirror section
3.0.0	October 2025	Updated new screenshots
3.0.1	January 2026	Workaround for AWS creds

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